

ENGINEERING PORTFOLIO SAMPLE



DAREN SOMBILON

TABLE OF CONTENTS



Objective	3
Software Expertise and Industries Served	4
Inventor Work Procedure	5-10
Standard Used	12
Modelling Technique	12



OBJECTIVE

With over nine years of experience in engineering design, technical drafting, and project coordination, I am pursuing a role as a Engineering Draftsperson with a focus on technical drafting, and process documentation. I offer strong proficiency in CAD software such as AutoCAD, Inventor, Onshape and Solidworks, alongside hands-on experience in 3D modeling, sheet metal design and technical production drawing. My goal is to contribute to the development of accurate, efficient process designs and support seamless project execution, cross functional coordination, and overall operational efficiency.

SOFTWARE EXPERTISE

INDUSTRIES SERVED

AUTOCAD

From aluminum and glass fabrication to cement construction, maintenance, and site development planning. I collaborate with the team to bring innovative solutions by creating precise 2D and 3D designs tailored to development needs



INVENTOR

My experience in creating 2D drawings and enhancing 3D automation processes for designing spools, handrails, CNC files, and cement plant structures has had a significant impact on standardization.



SOLIDWORKS

Skilled in sheet metal design for industries such as mining and electrical engineering, with expertise in wiring enclosures, ergonomic brackets, and automated weldments for structural and residential applications.

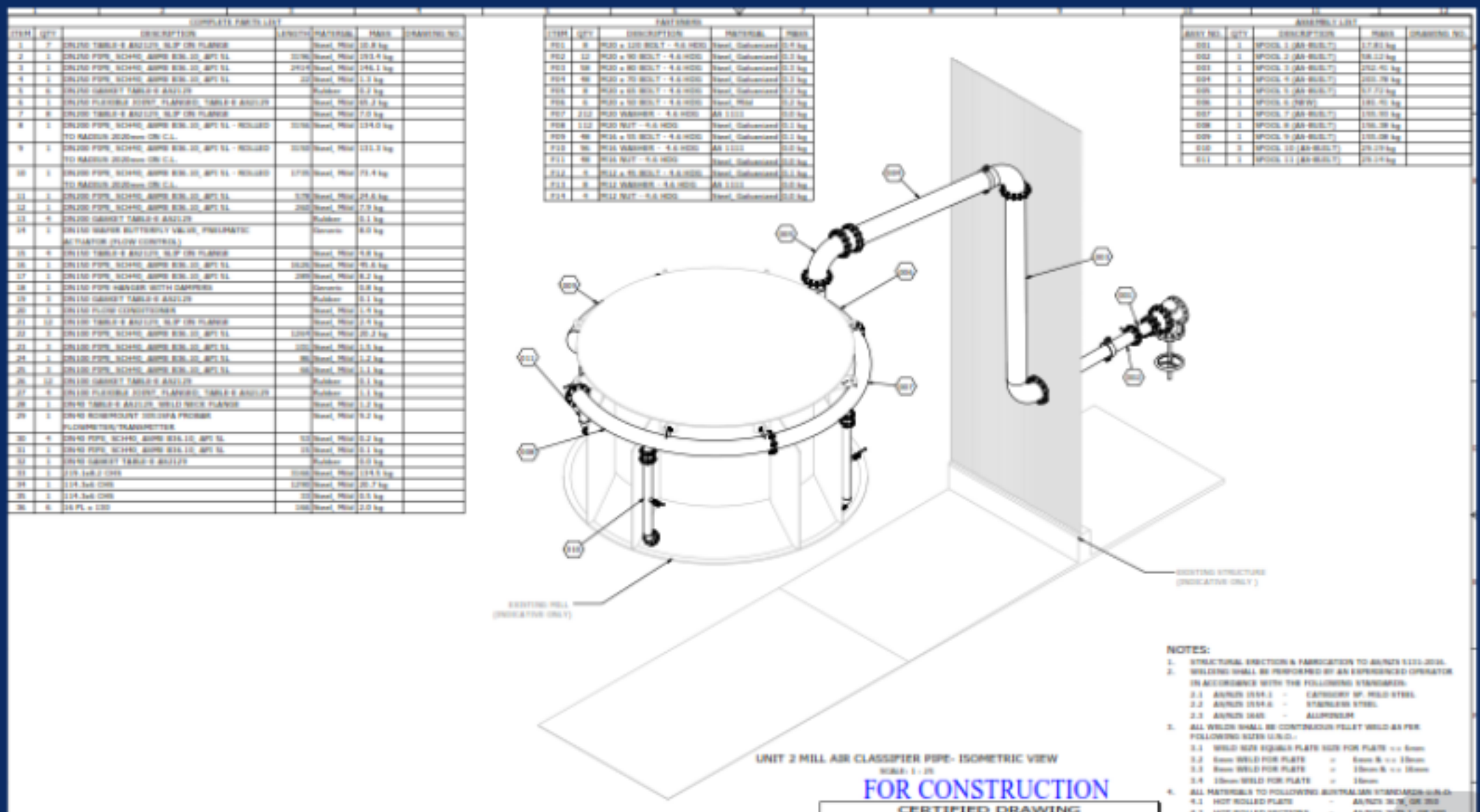


METHODS FOR DRAWING

1. Drawing Sheet Setup

2. Index Sheet and Notes Sheet for large project is commonly required

3. General Arrangement Drawing

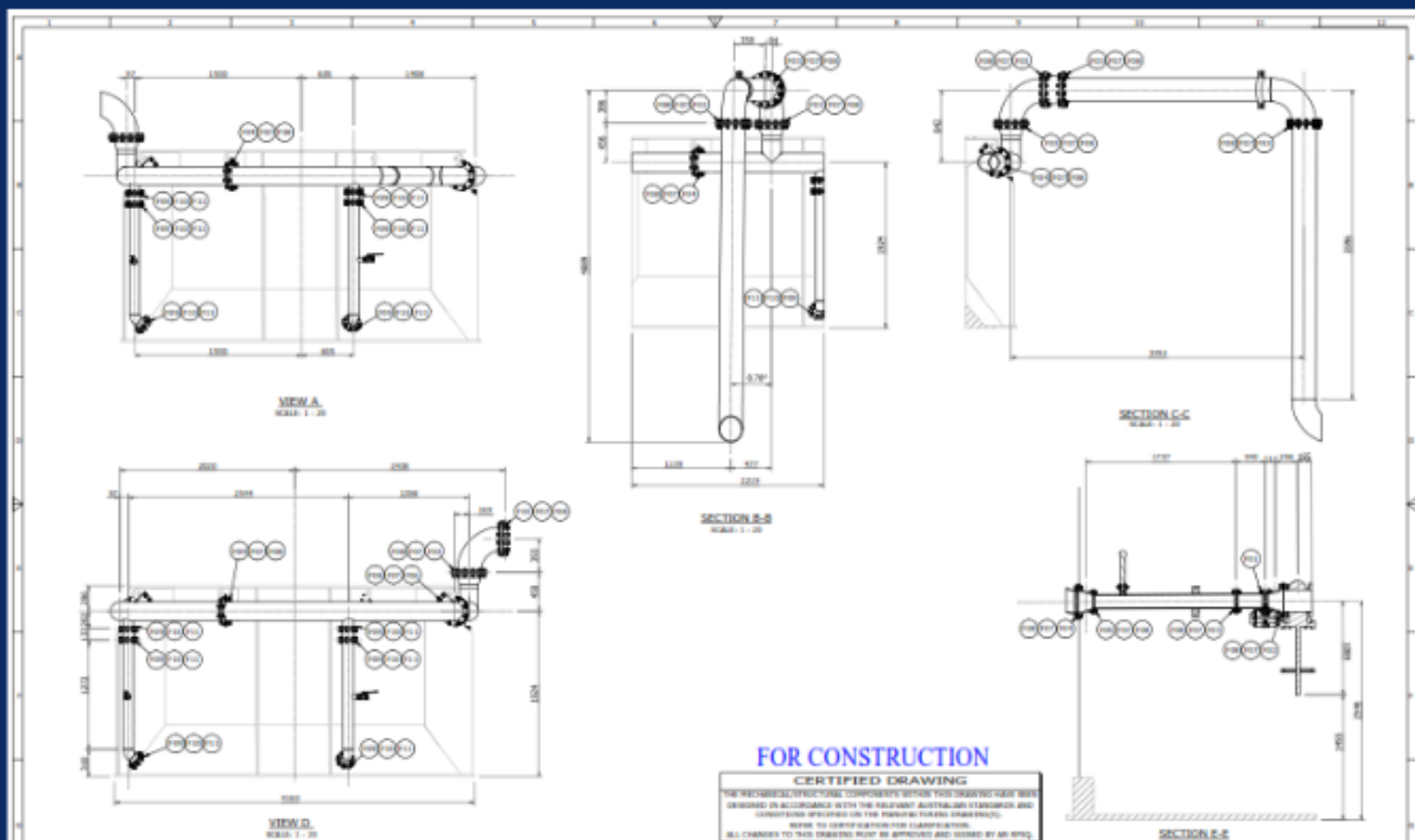
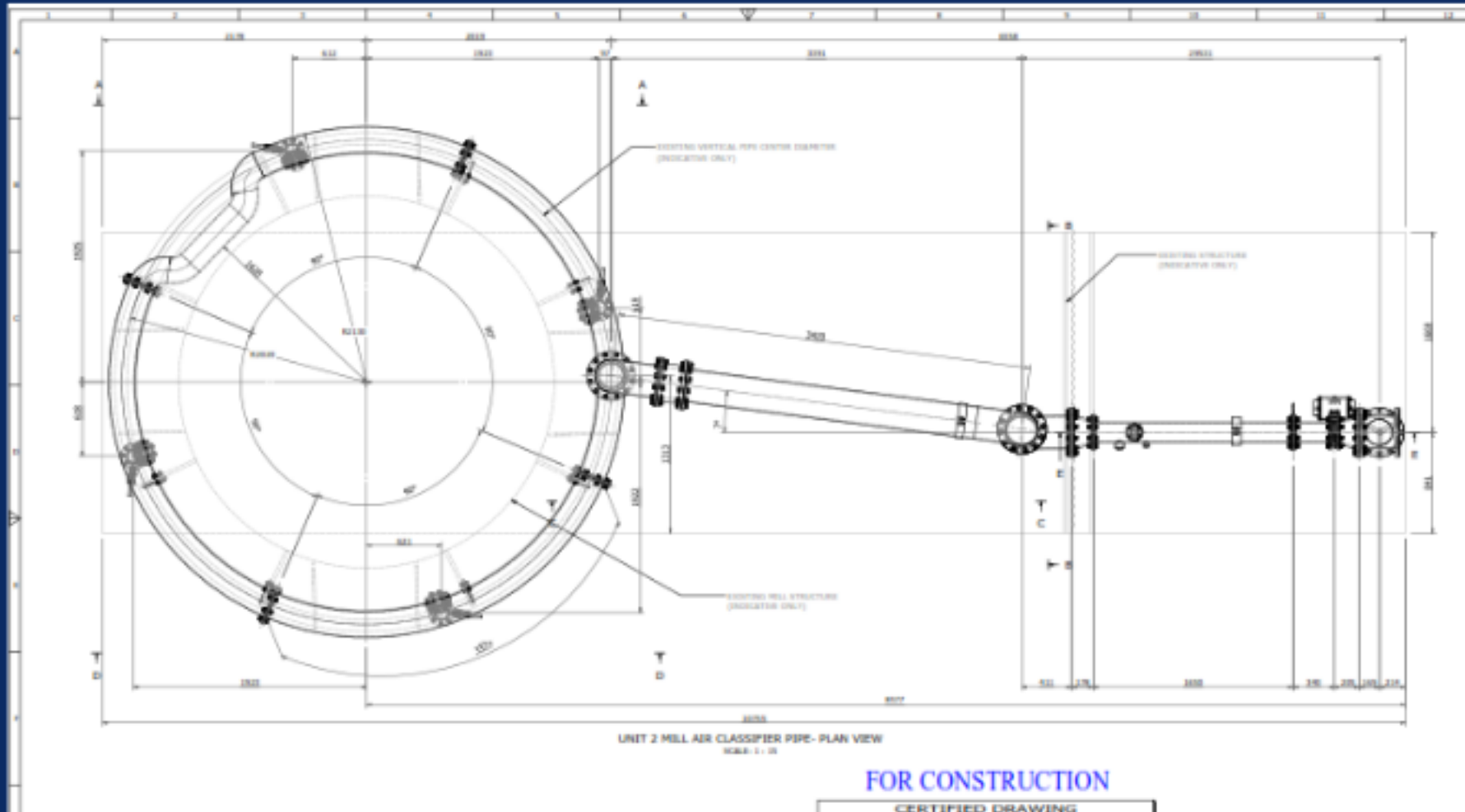


INVENTOR WORK PROCEDURE

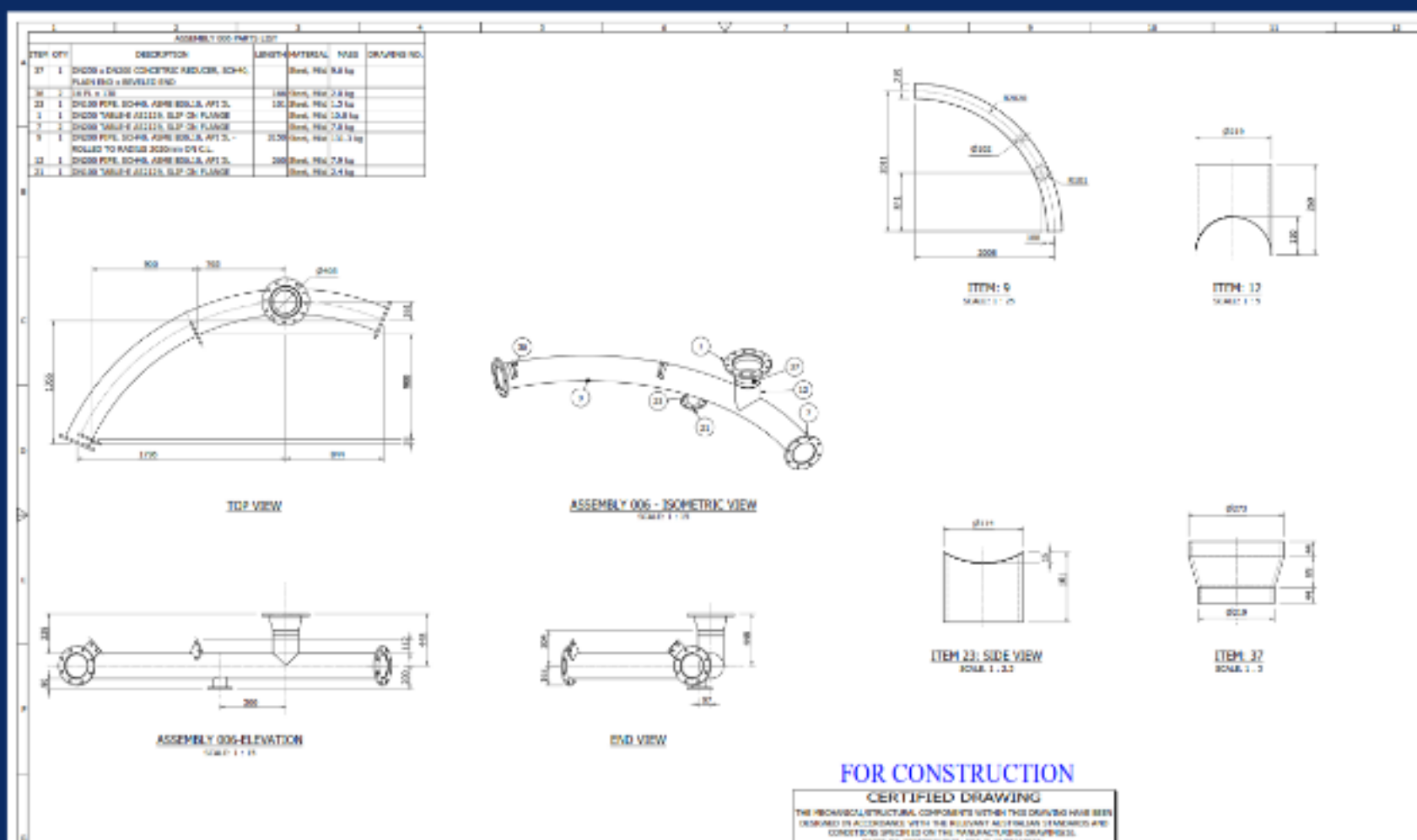
METHODS FOR DRAWING

DRAWING SET SEQUENCE AND LAYOUT

4. Set-out Drawing -Site



5. Assembly Detail Drawings & Part Drawings

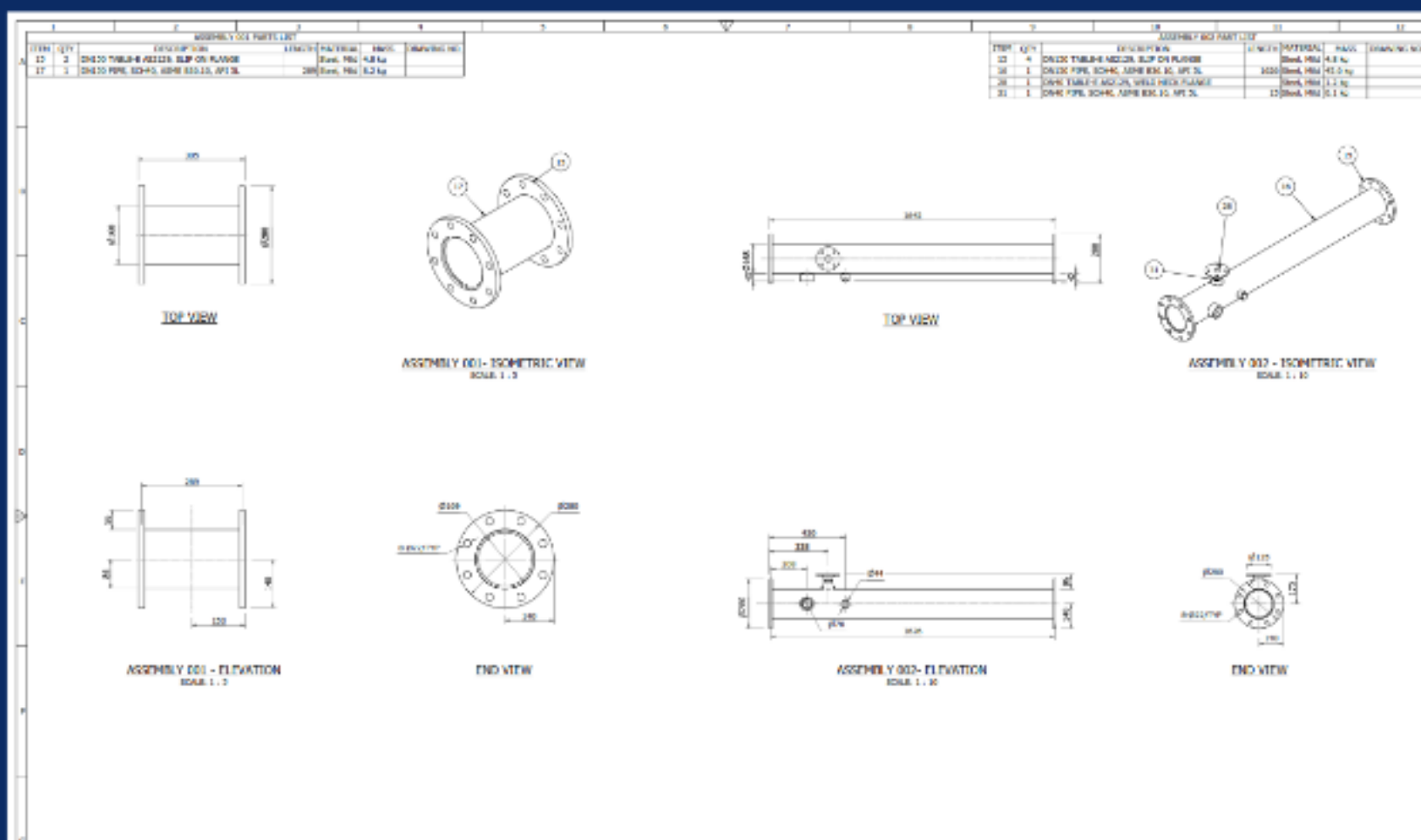
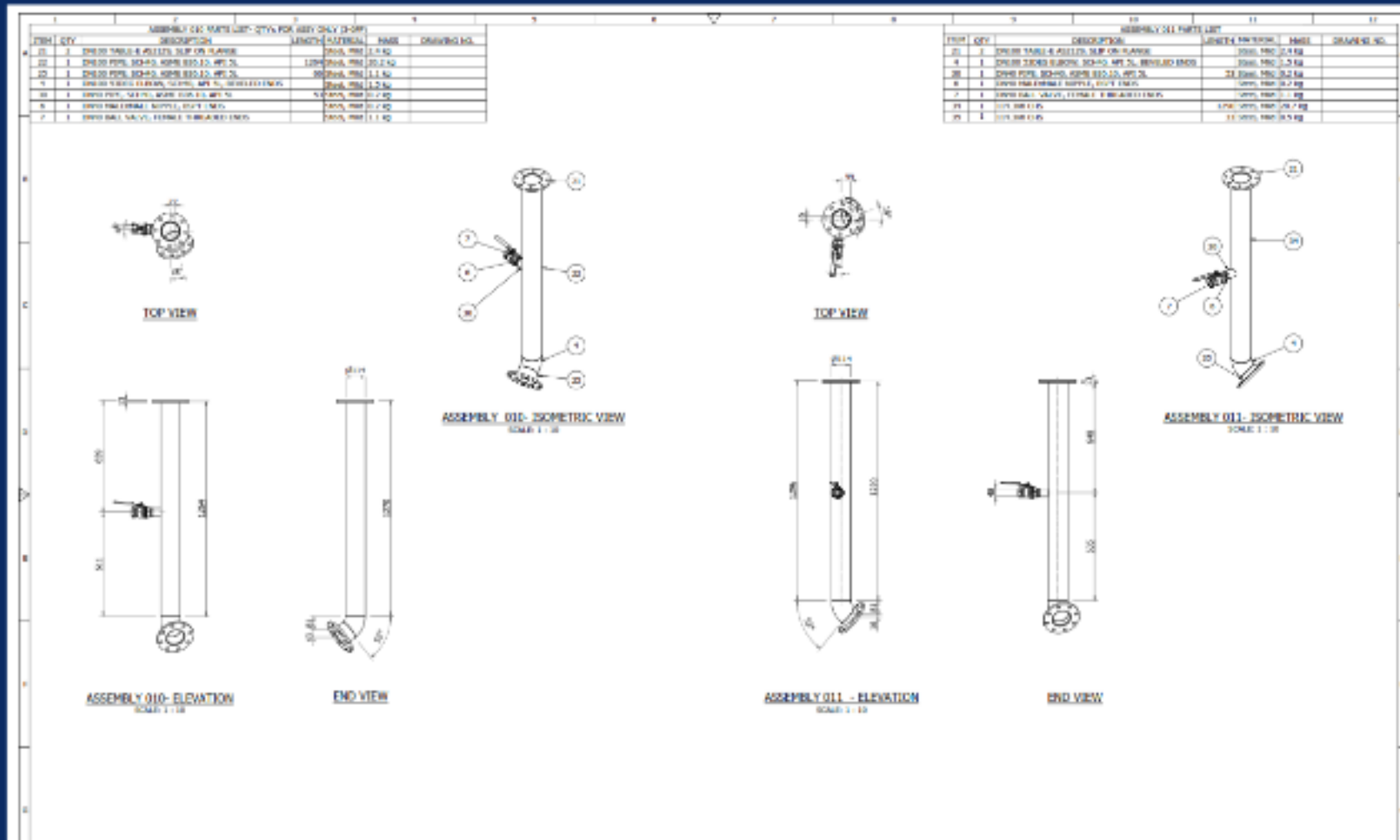


INVENTOR WORK PROCEDURE

METHODS FOR DRAWING

DRAWING SET SEQUENCE AND LAYOUT

5. Assembly Detail Drawings



INVENTOR WORK PROCEDURE

METHODS FOR DRAWING

PRESENTATION OF ARRANGEMENT AND SETOUT DRAWINGS

This method enable to clarify or enhance the specific projection, arrangement view, list of tables needed, or even the color projection

1. Existing components and structure - colour
2. Assembly List - table
3. Parts List - table
4. Fasteners - table
5. Detailing of Sub Assemblies and Parts - projection, arrangement view
6. Labelling of Parts - naming standard.

DRAWING CONTENT

1. View labels
2. Dimensioning
3. Sketch Symbols
4. Drawing Notes

DRAWING TITLE BLOCK TEMPLATES

1. Drawing Title Block Selection
2. Drawing Title Block Data Entry and Entry

CREATING DXF FILES

1. Standard Plates, Sheet Metal or Flat Bar
2. Pressed Plate

EXPORTING INVENTOR DRAWINGS TO DWG FORMAT

1. Export to DWG Format

ILOGIC

1. External Rules and Forms



STANDARD USED

AUSTRALIAN STANDARDS...

1. AS 1100 – Technical Drawing
2. AS 1100.101 – Technical Drawing – General Principles
- 3.) AS 1100.201 – Technical Drawing – Mechanical Drawing
- 4.) AS 1100.301 – Technical Drawing – Architectural Drawing
- 5.) AS 1100.401 – Technical Drawing – Engineering Survey and Engineering Survey Design Drawing
- 6.) AS 1100.501 – Technical Drawing – Structural Engineering Drawing



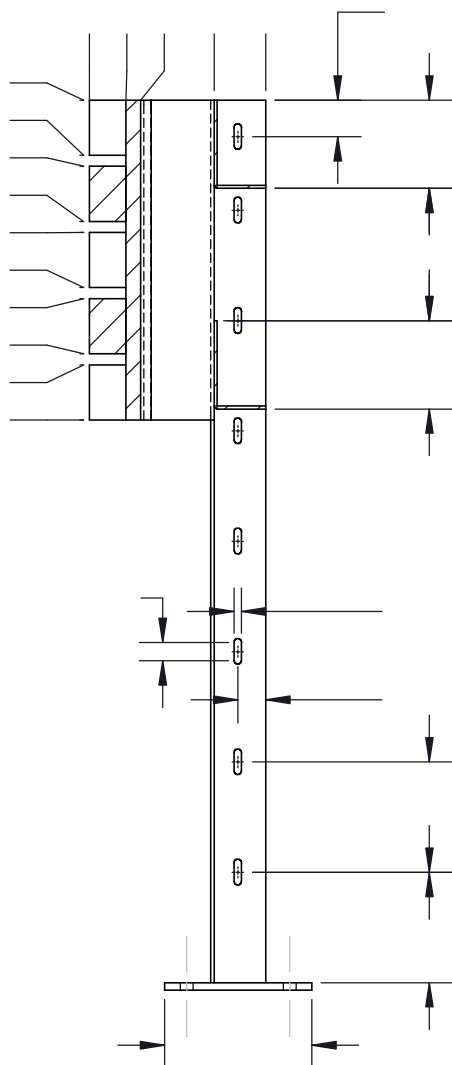
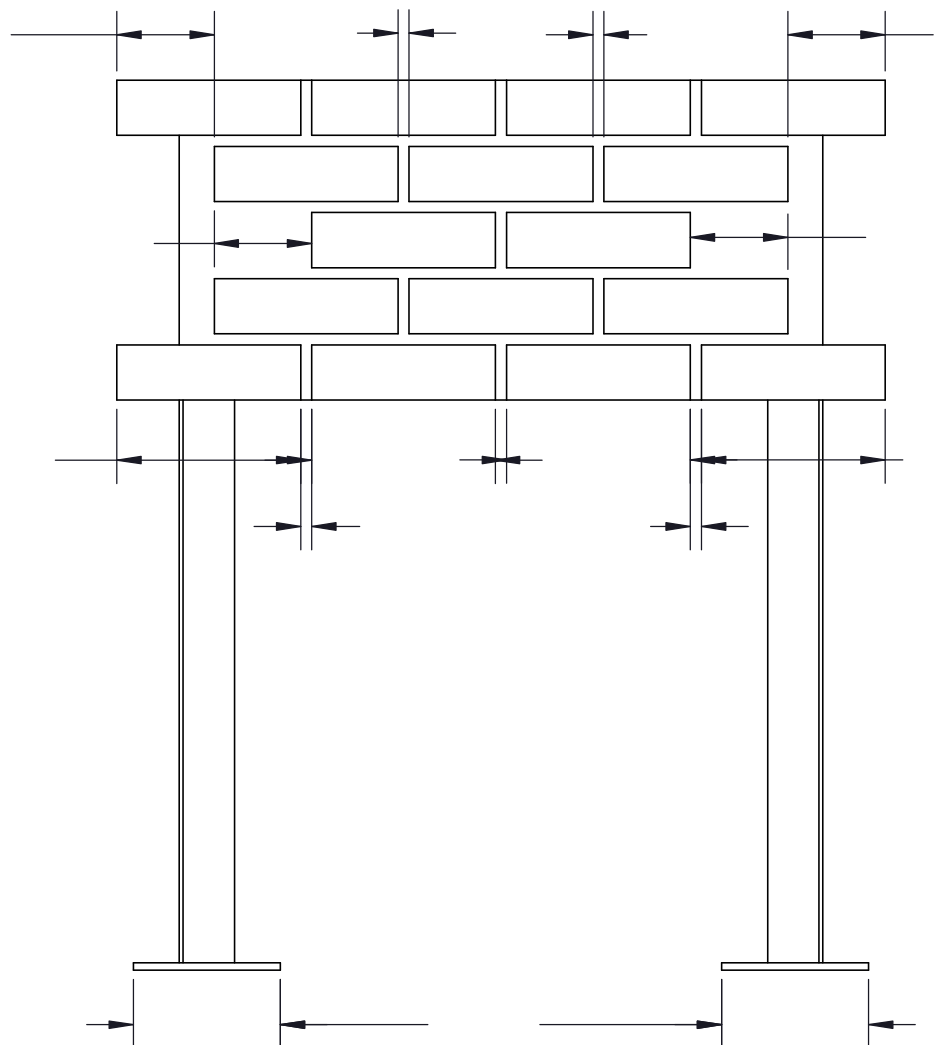
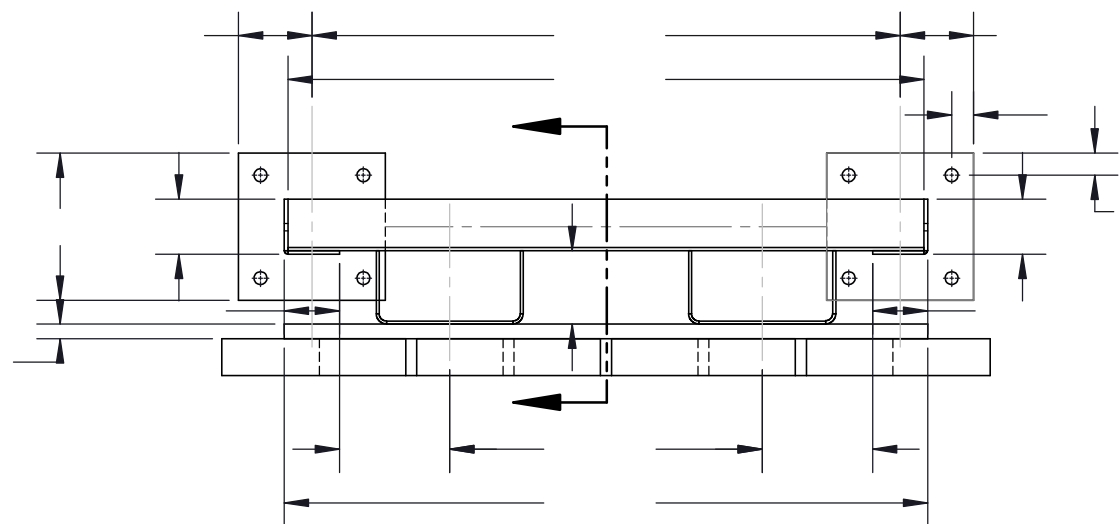
MODELING TECHNIQUE

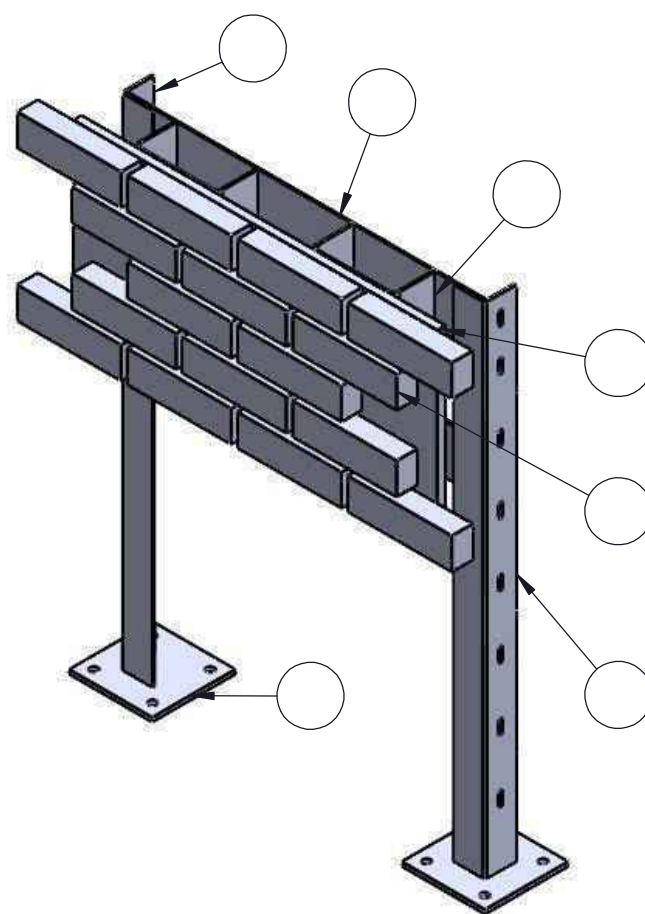
MODELLING WITH THE MASTER PART METHODOLOGY

1. Using Top-Down design approach using Master Part Methodology.



SOLIDWORKS
SHEET METAL & WELDMENTS
PORTFOLIO SAMPLE

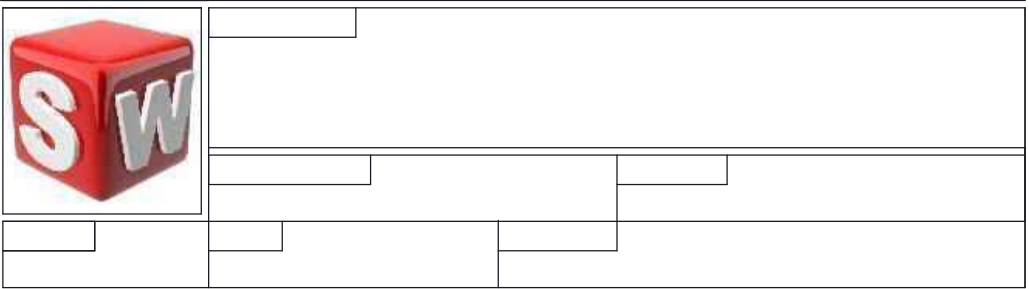
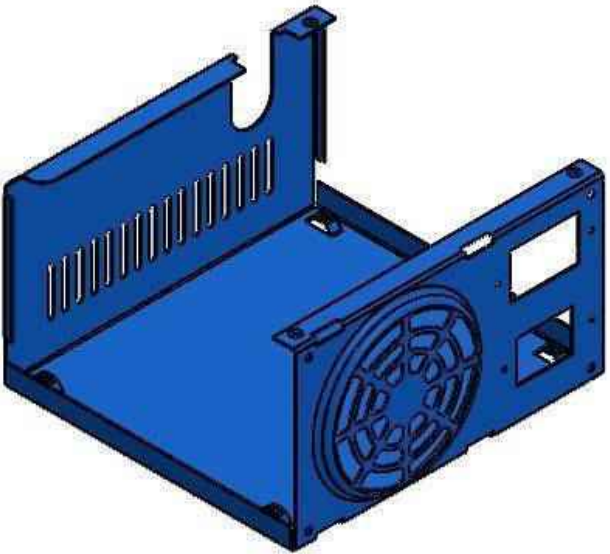
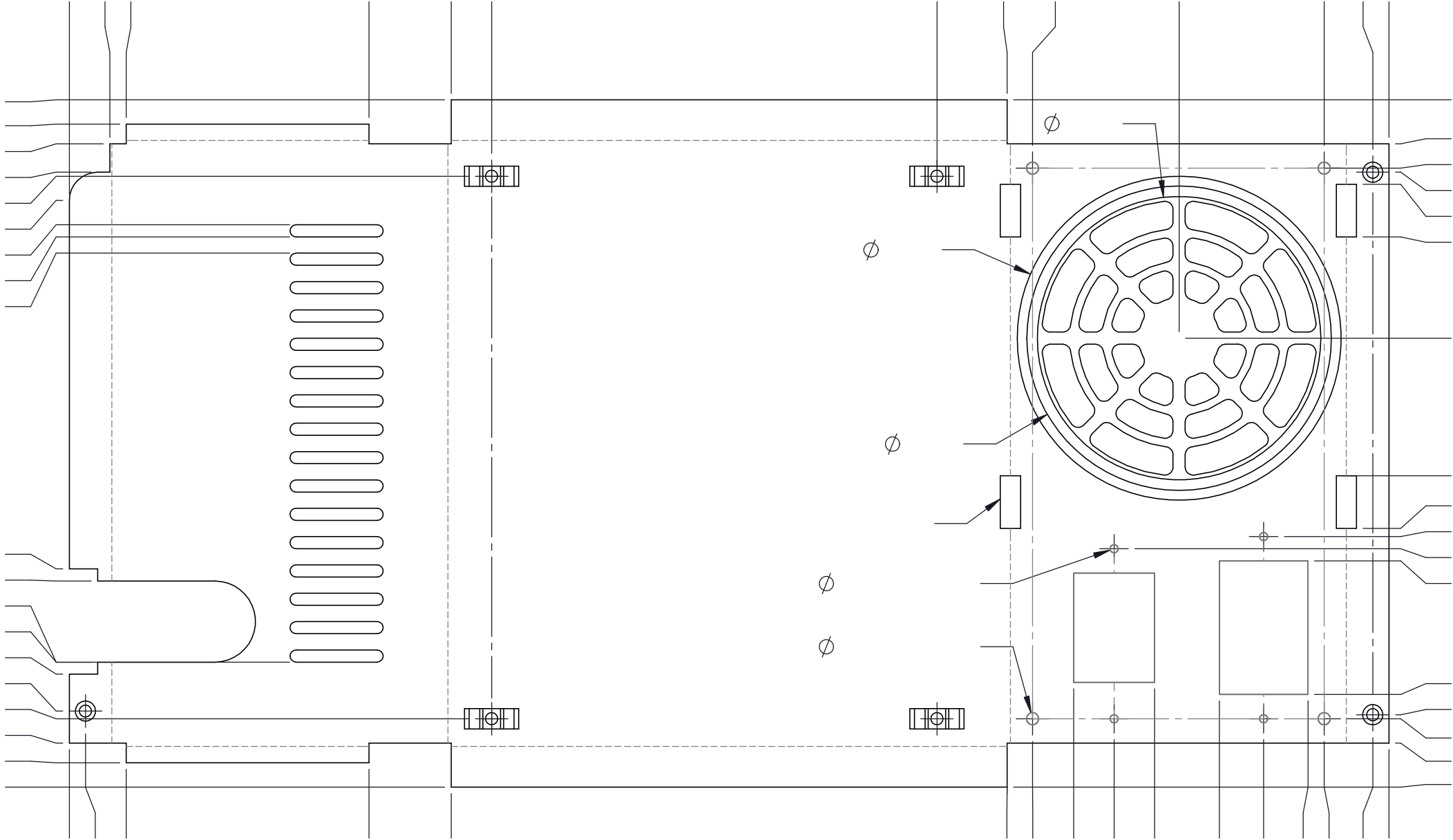




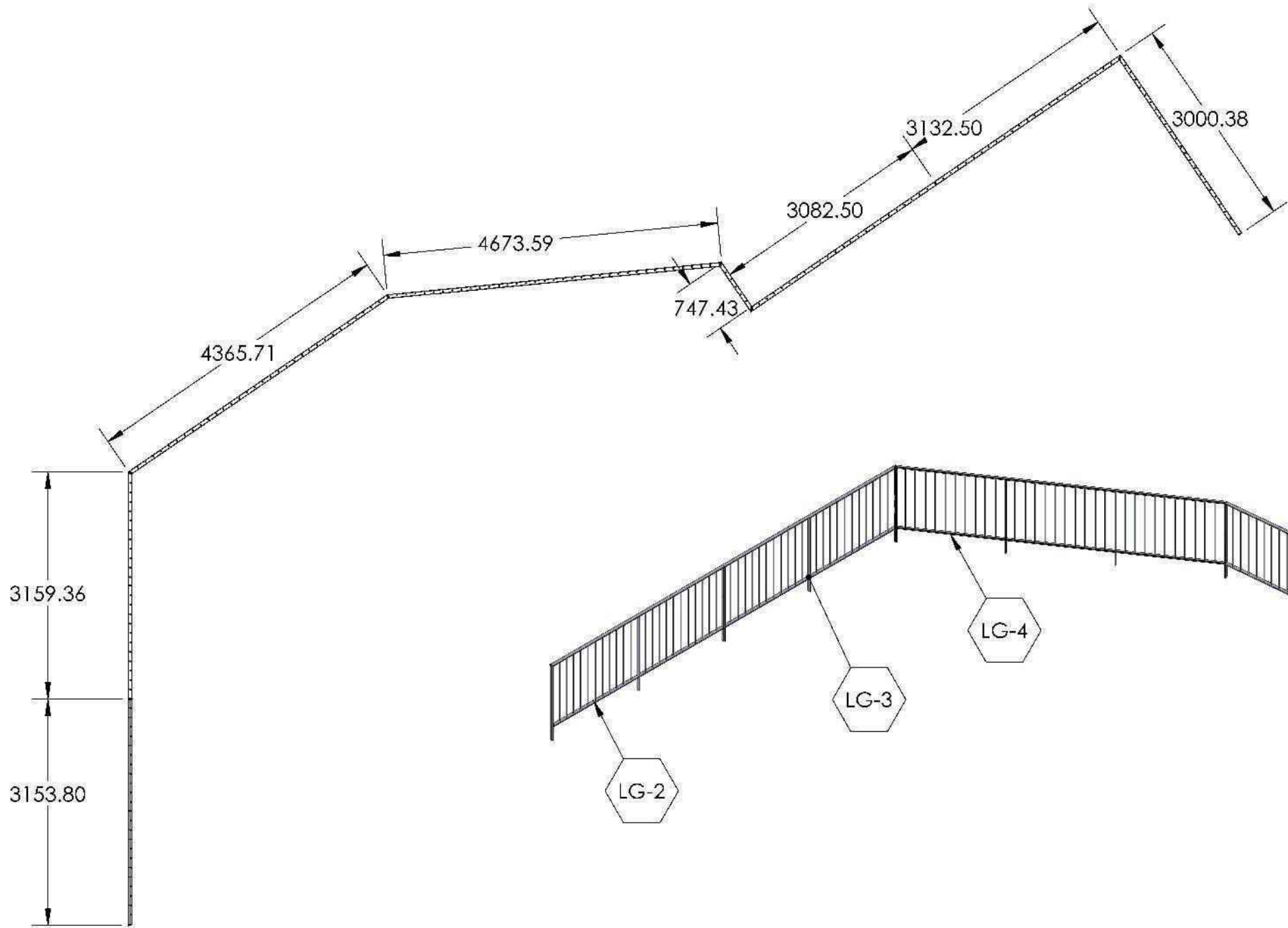
[illegible][illegible]

FLAT PATTERN
WITH DXF UPON REQUEST



EXAMPLE OF GENERAL ASSEMBLY WITH WELDMENTS APPLICATION IN SOLIDWORKS

ASSEMBLY NO.	DESCRIPTION	QTY.
LG-2	LOWER GROUND BALUSTRADE 2	1
LG-3	LOWER GROUND BALUSTRADE 3	1
LG-4	LOWER GROUND BALUSTRADE 4	1
LG-5	LOWER GROUND BALUSTRADE 5	1
LG-6	LOWER GROUND BALUSTRADE 6	1
LG-7	LOWER GROUND BALUSTRADE 7	1
LG-8	LOWER GROUND BALUSTRADE 8	1
LG-9	LOWER GROUND BALUSTRADE 9	1



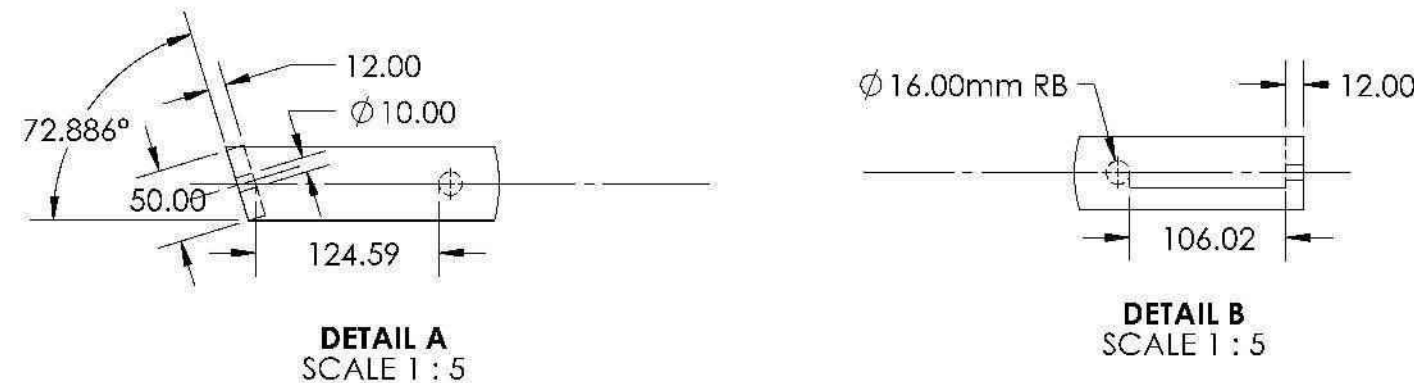
TOP VIEW
SCALE 1 : 70

ISOMETRIC VIEW: LOWER GROUND BALUSTRADE
SCALE 1 : 65

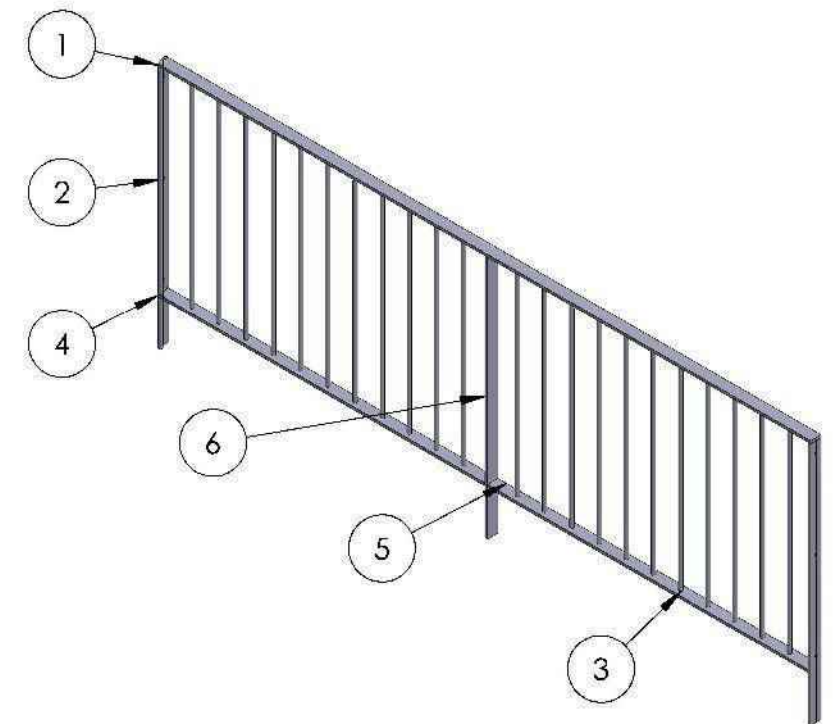
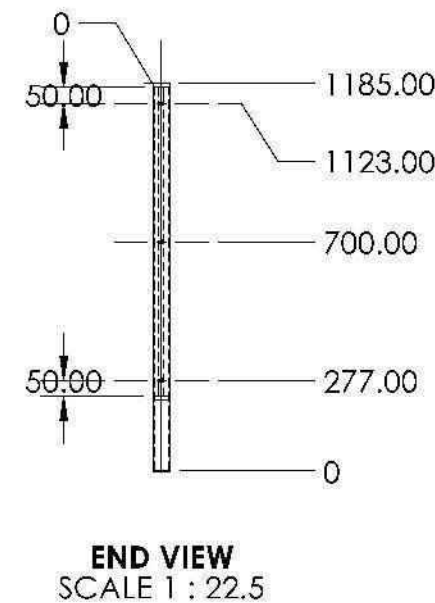
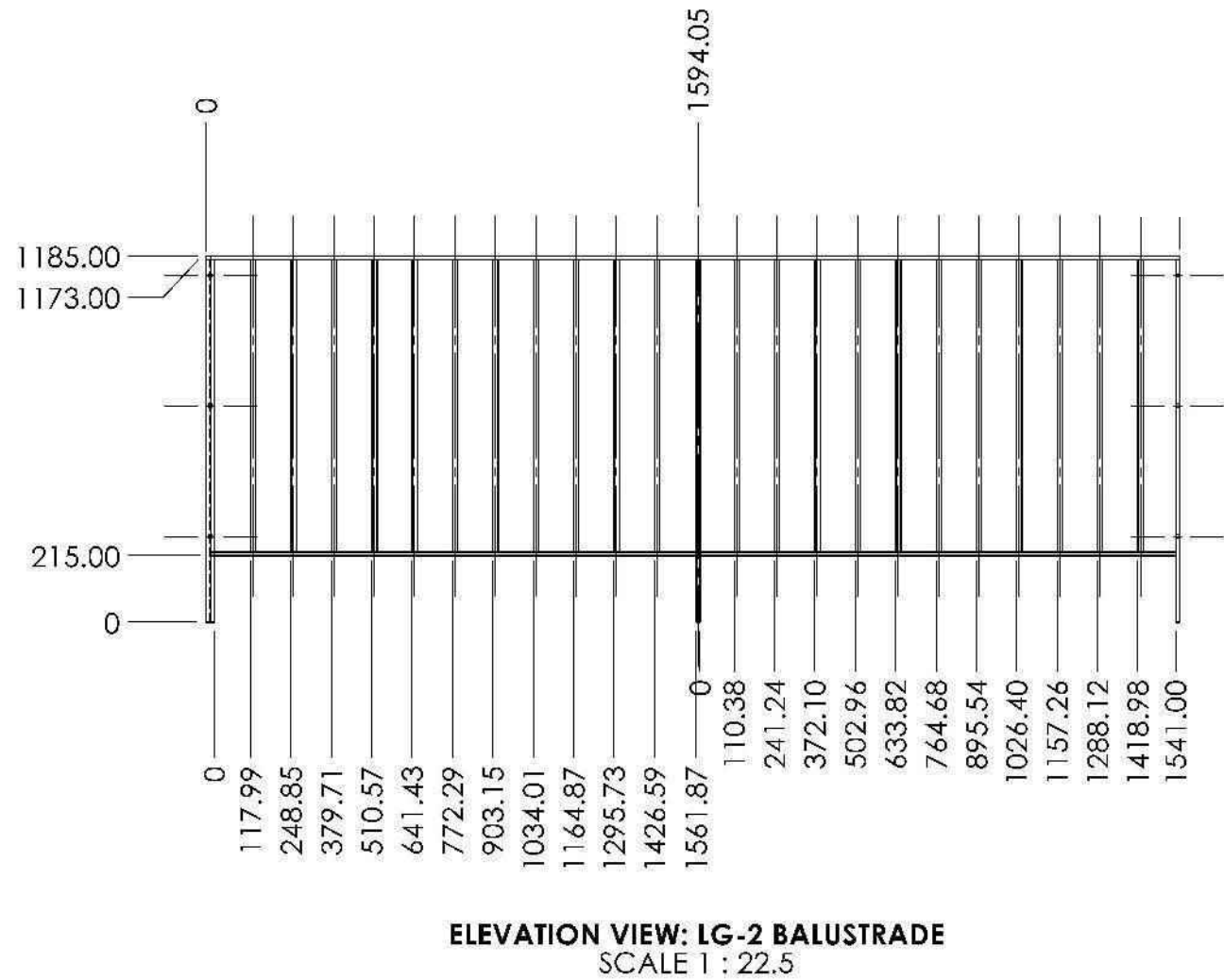
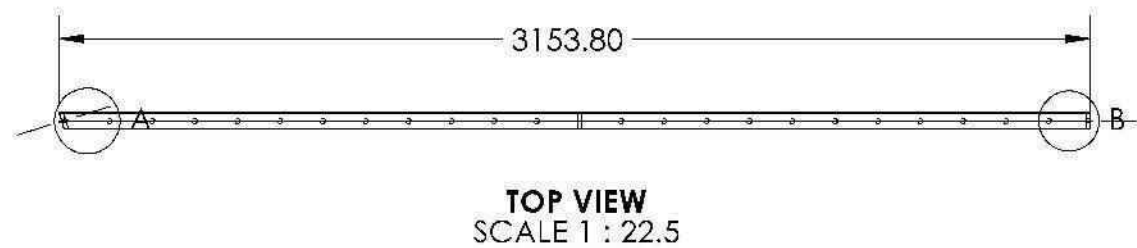


DESCRIPTION:	
GENERAL ASSEMBLY LOWER GROUND BALUSTRADE	
ITEM NUMBER:	STAGE:
LG-2, LG-3, LG-4, LG-5, LG-6, LG-7, LG-8, LG-9	ISSUED FOR MANUFACTURING APPROVAL
JOB NO.:	DATE:
	03/03/2025

EXAMPLE OF SUB-ASSEMBLY DRAWING INVOLVING WELDMENTS



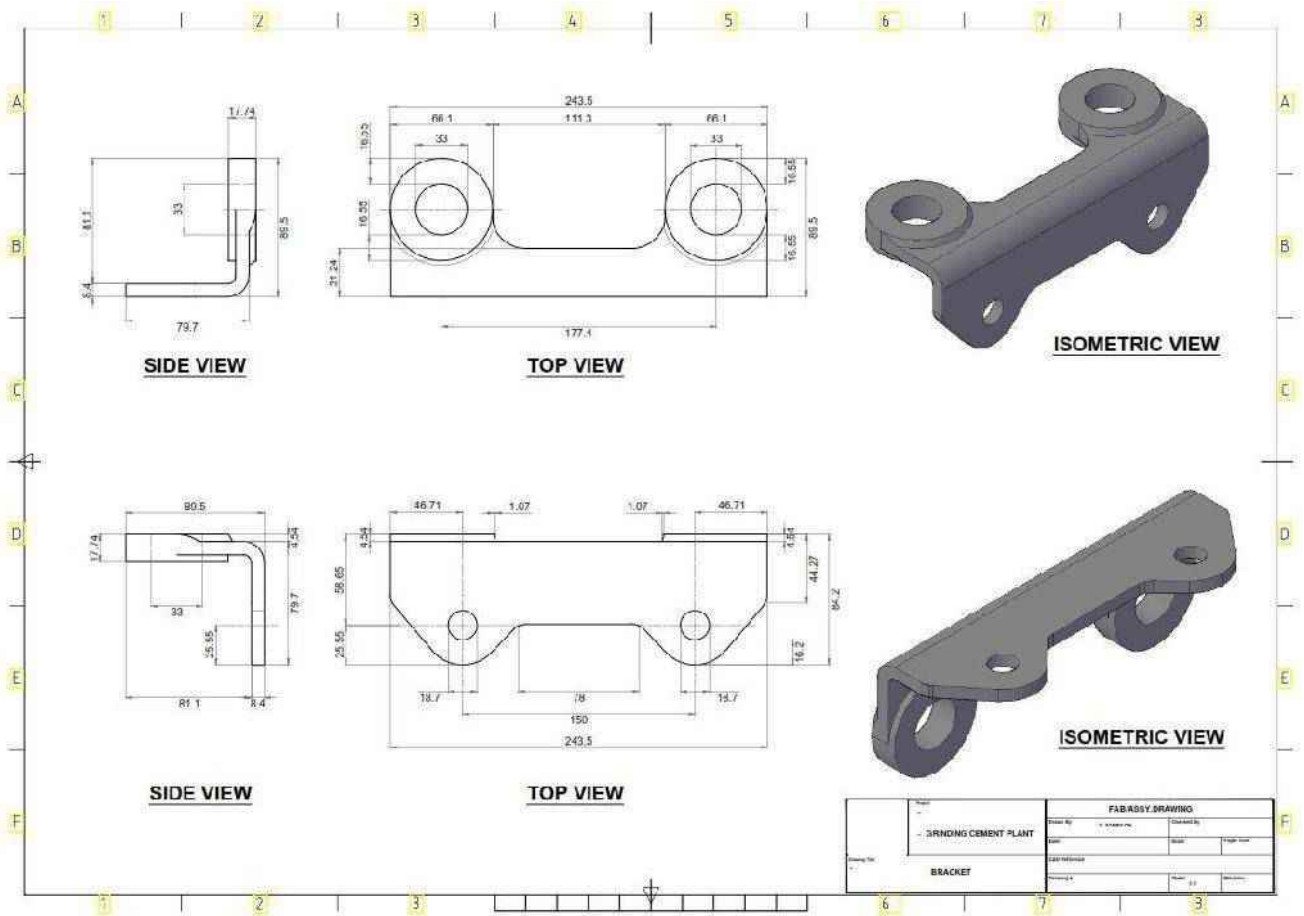
ITEM NO.	QTY.	DESCRIPTION	LENGTH
1	1	15 x 12mm FB (EX4032)	3153.8
2	2	15 x 12mm FB (EX4032)	1173
3	22	Ø16mm RB (EX6002)	946
4	1	15 x 12mm FB (EX4032)	1576.170332
5	1	15 x 12mm FB (EX4032)	1541
6	1	15 x 12mm FB (EX4032)	1173

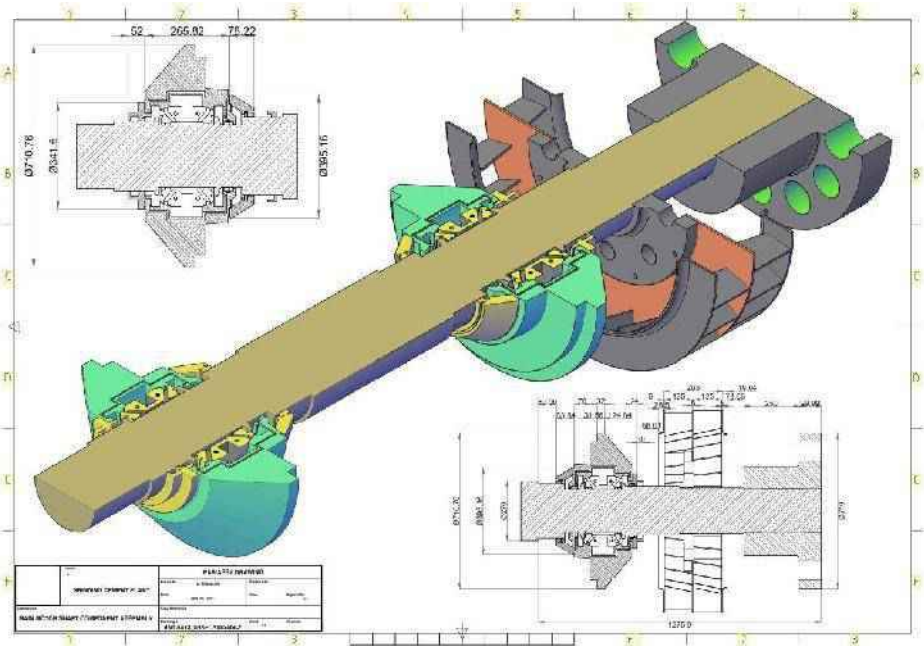
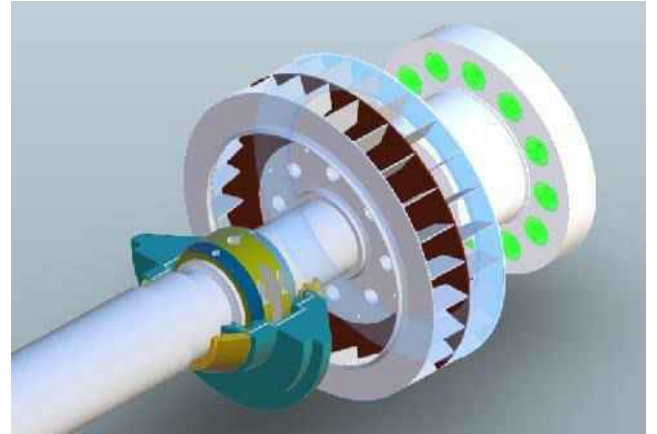
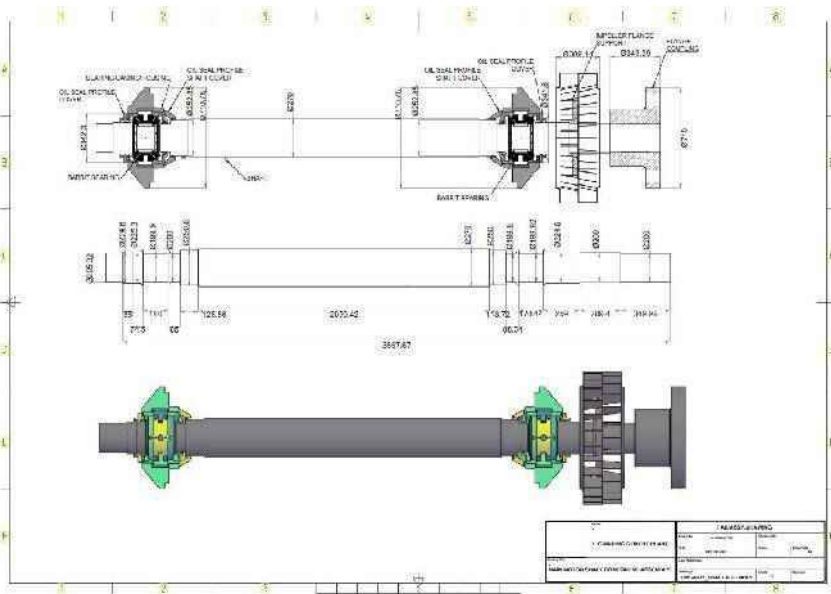


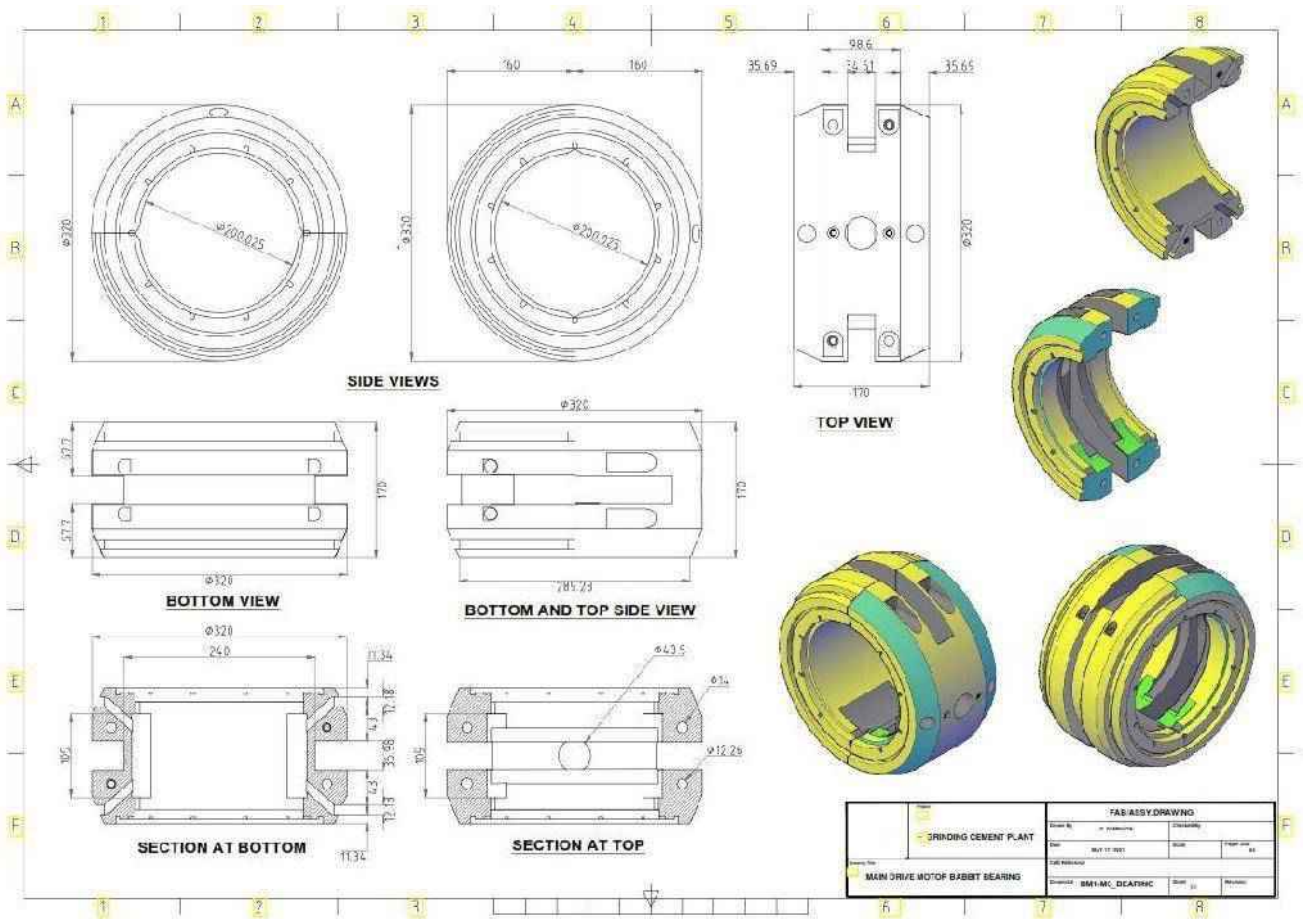
	DESCRIPTION:		
	LG-2 DETAIL LOWER GROUND BALUSTRADE		
JOB NO:	DATE:	WEBSITE:	STAGE:
03/03/2025	03/03/2025		ISSUED FOR MANUFACTURING APPROVAL

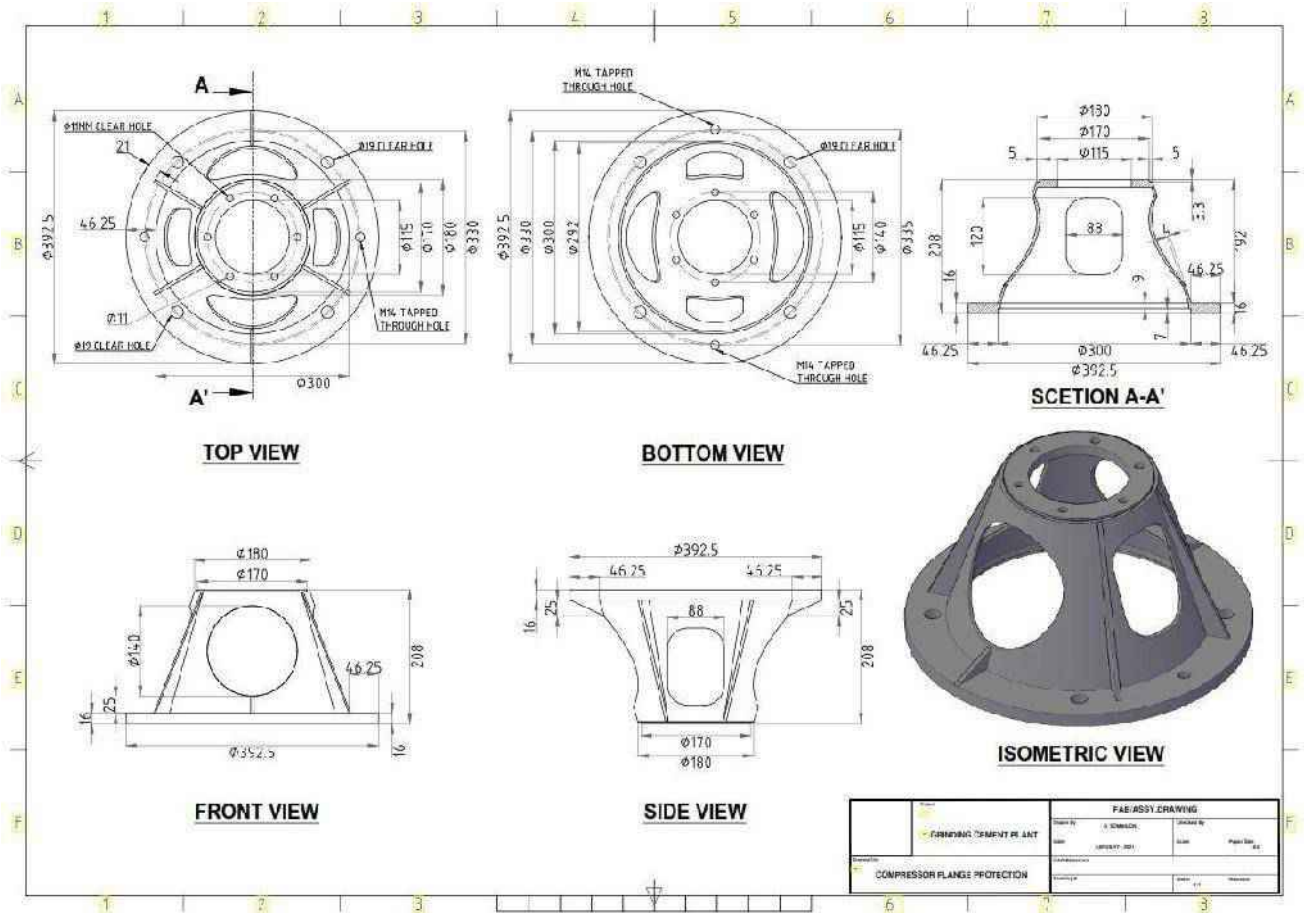
AUTOCAD

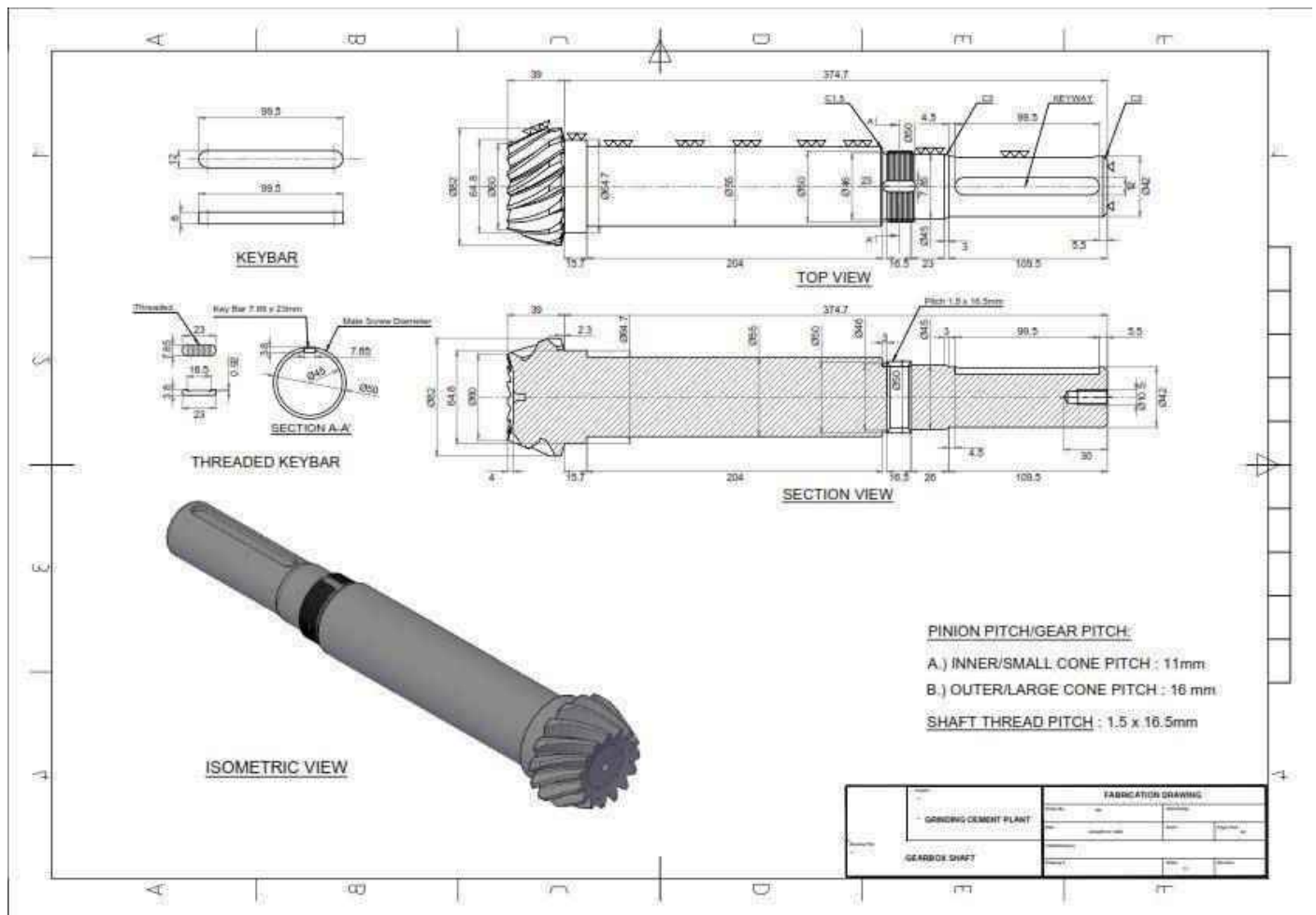
3D MODELING AND TECHNICAL DRAWINGS

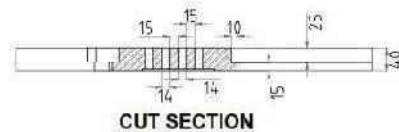
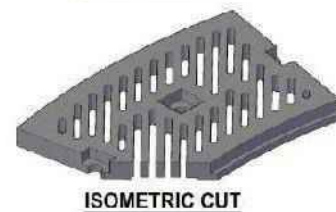
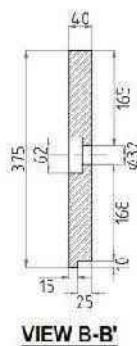
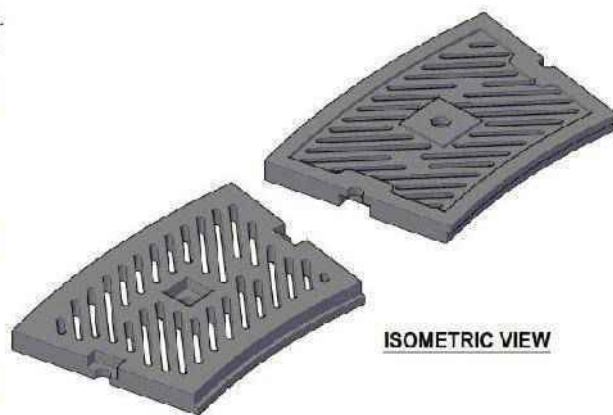
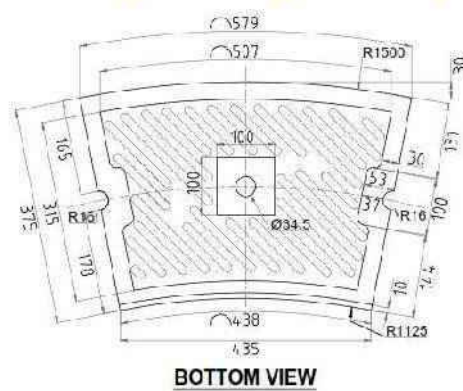
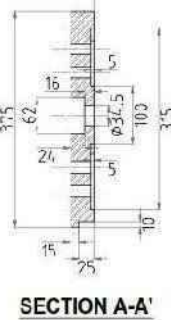
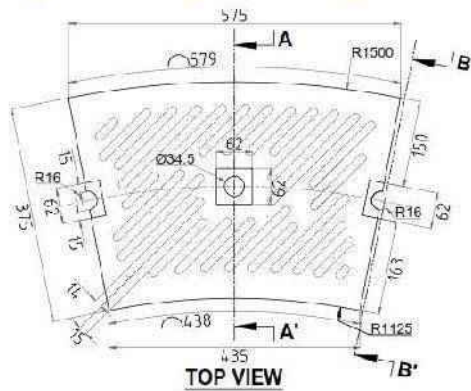






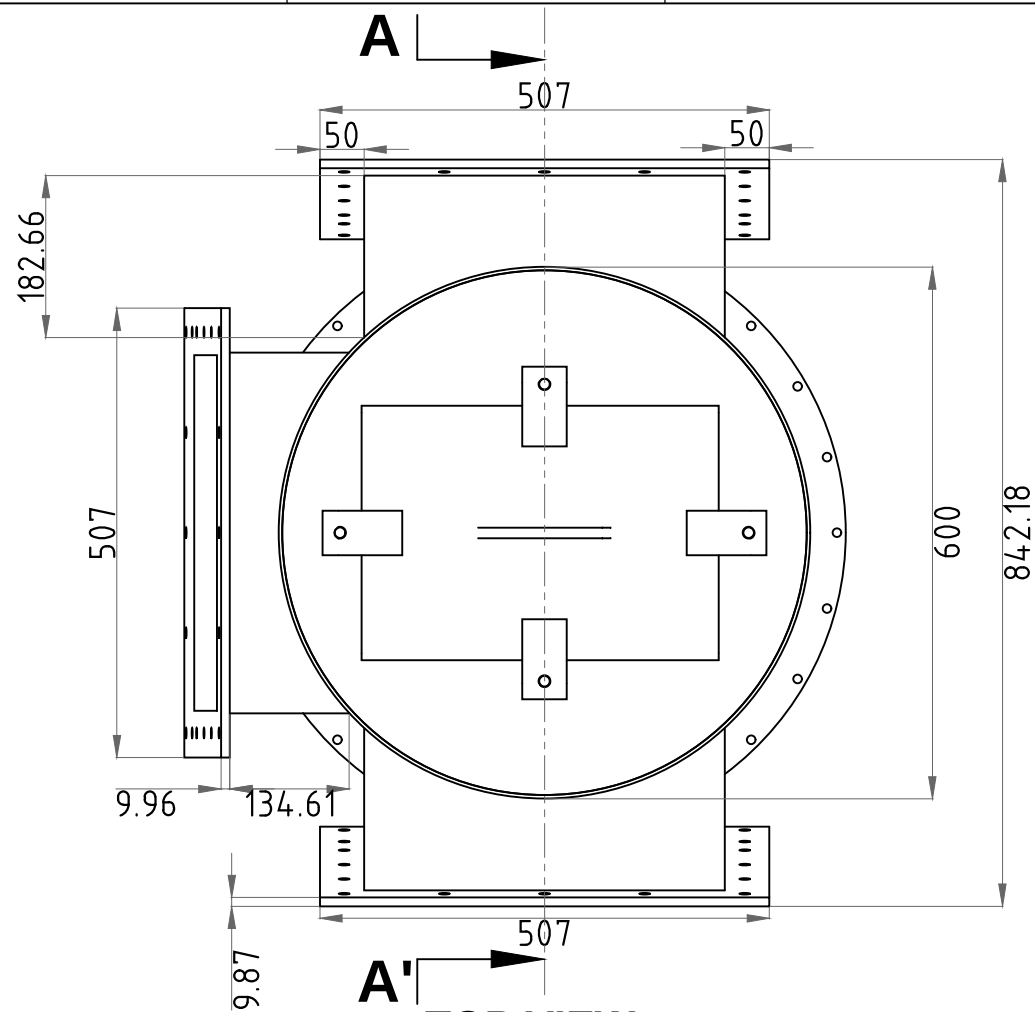




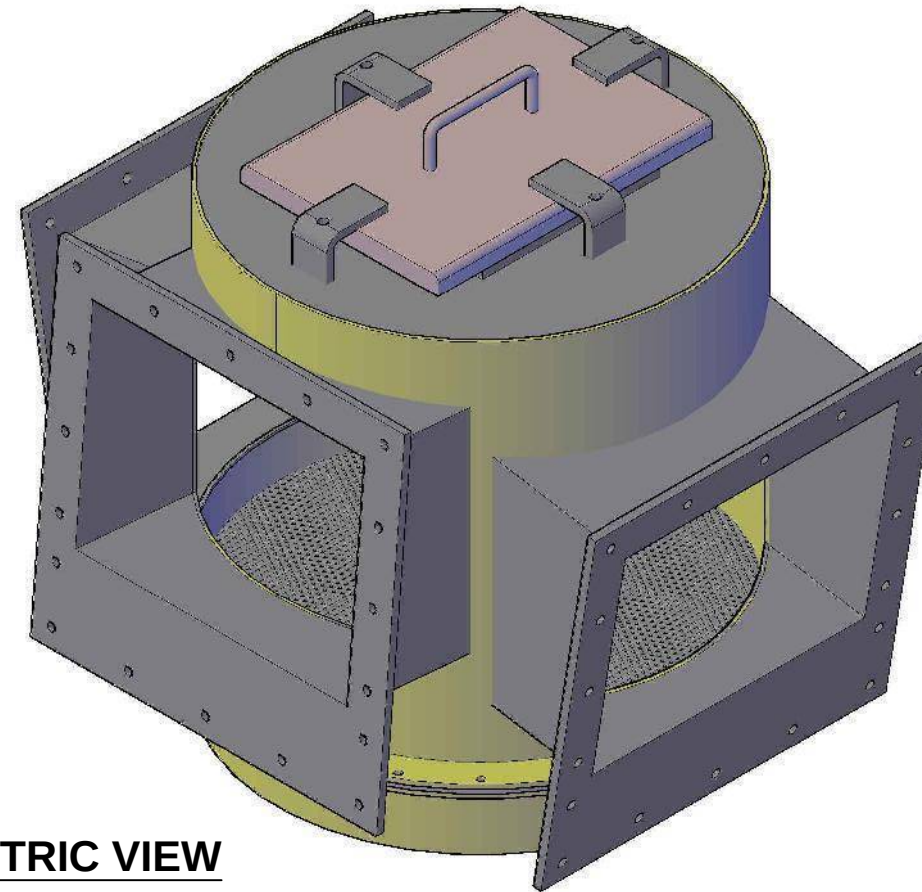


PART		PART ASSY. DRAWING			
GRINDING WHEEL FLANT		Drawn By	Checked By	Drawn Date	Check Date
BALL MILL 1 DISCHARGE LINER		DATE	DATE	DATE	DATE
OUTER RACE CHAMBER 1		DATE	DATE	DATE	DATE
		DATE	DATE	DATE	DATE

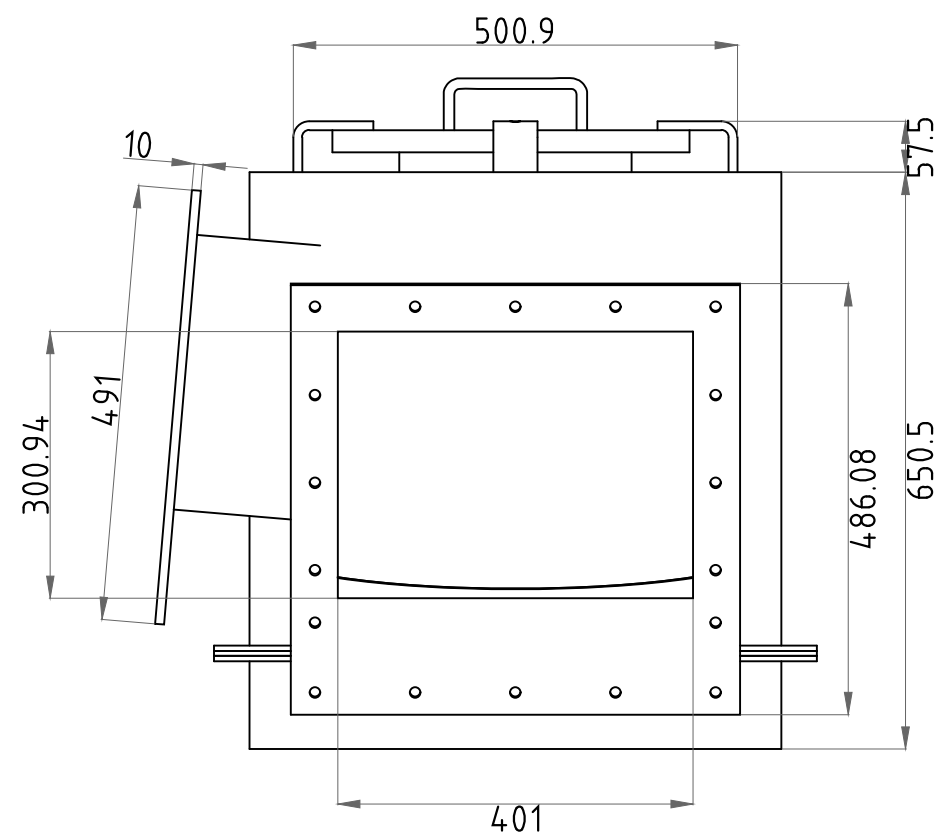




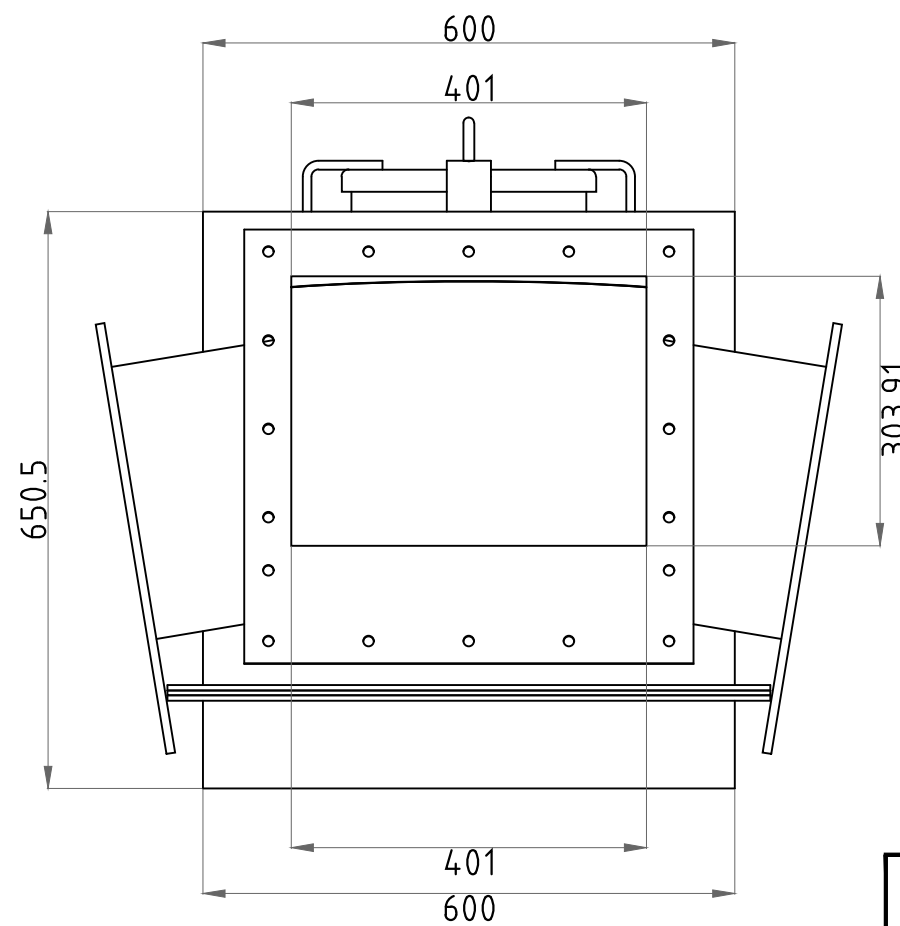
TOP VIEW



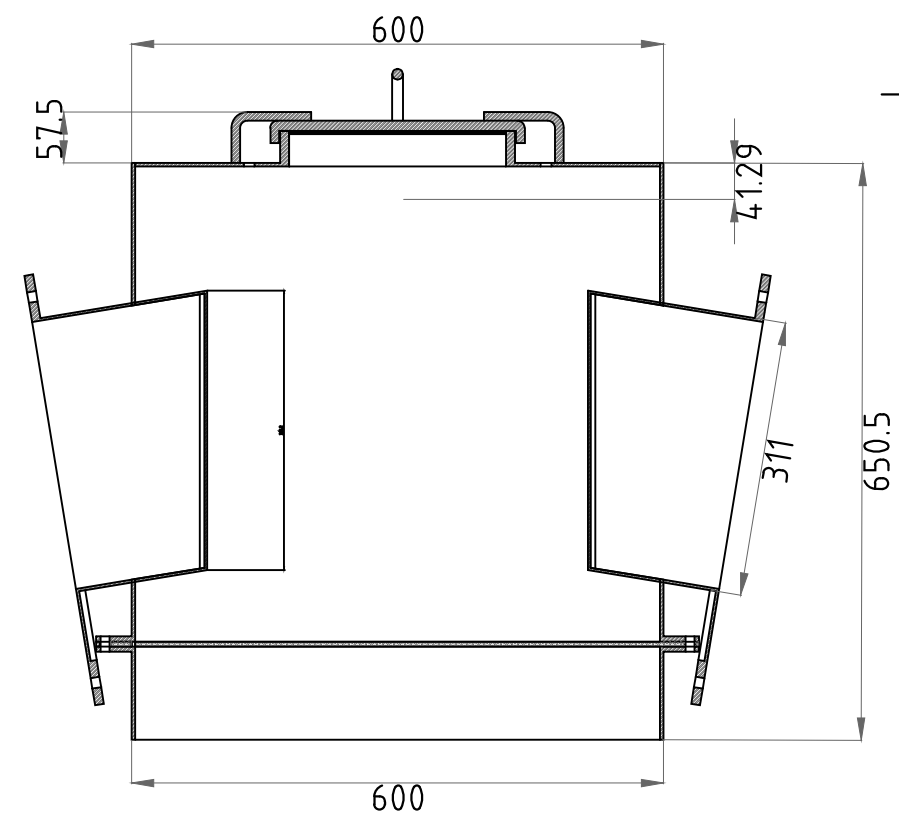
ISOMETRIC VIEW



FRONT VIEW

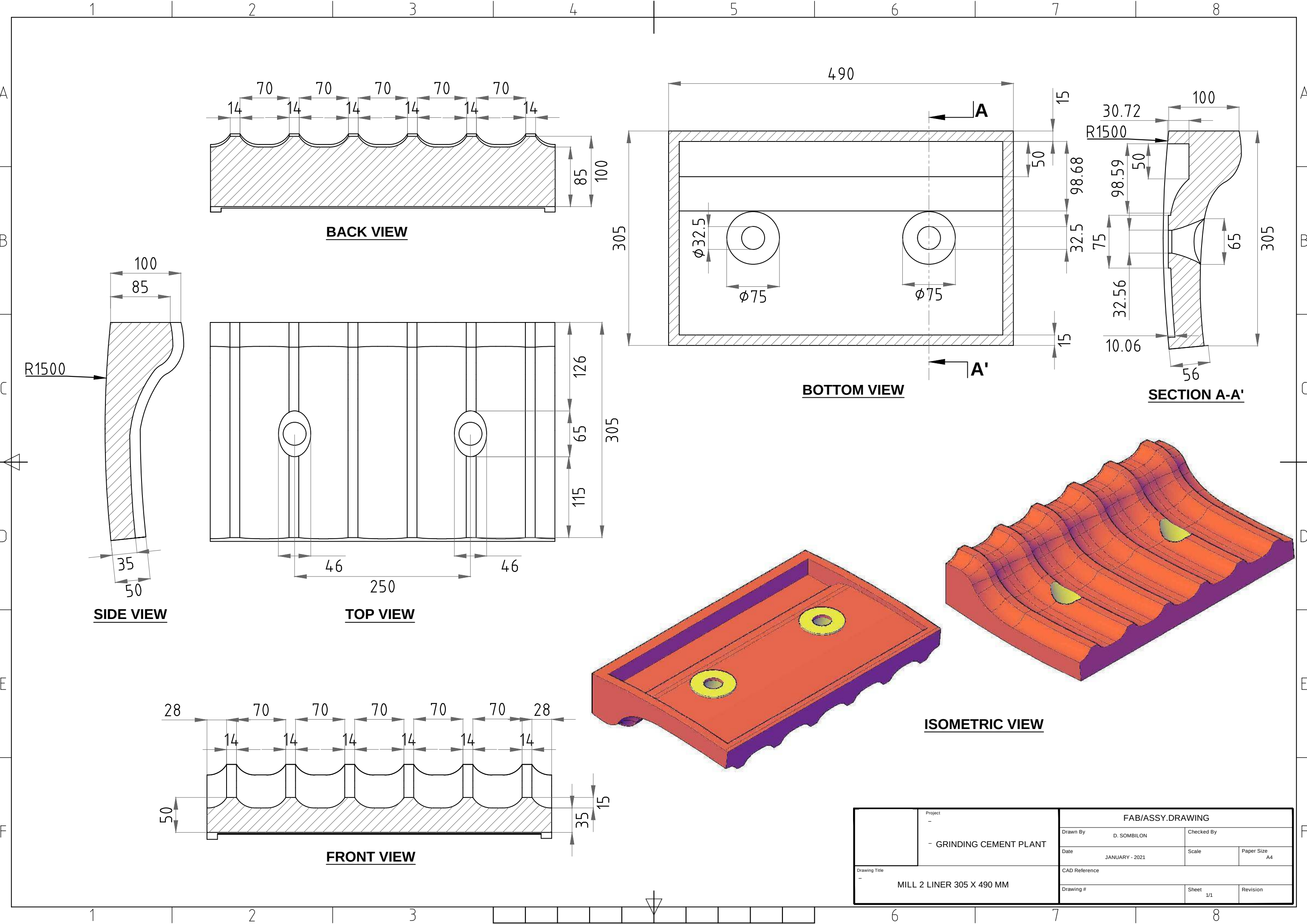


SIDE VIEW

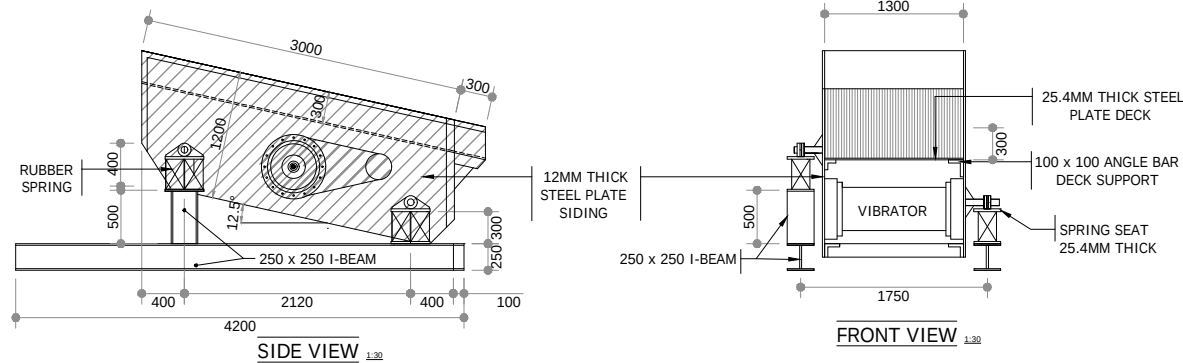
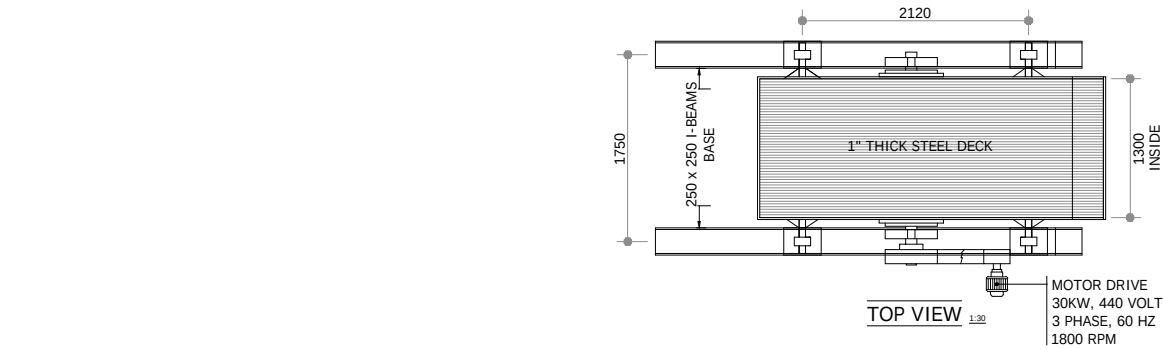


SECTION A-A'

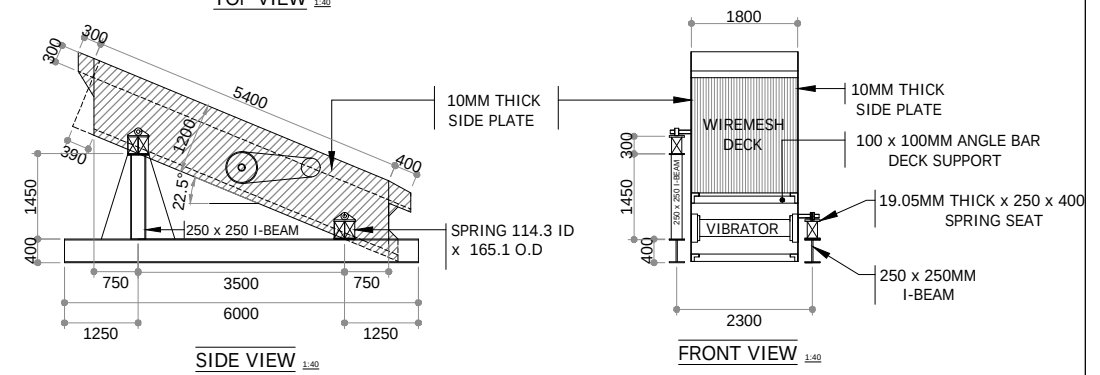
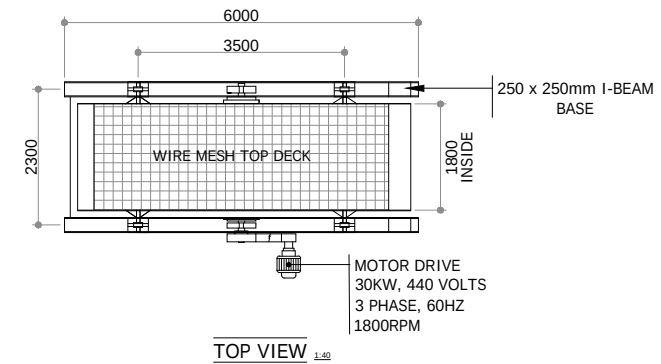
Project - - GRINDING CEMENT PLANT	FAB/ASSY.DRAWING			
	Drawn By D. SOMBILON		Checked By	
	Date DECEMBER - 2020		Scale	Paper Size A4
Drawing Title - AIRSLIDE DIVETER CONNECTOR ASSEMBLY	CAD Reference			
	Drawing #		Sheet 1/1	Revision



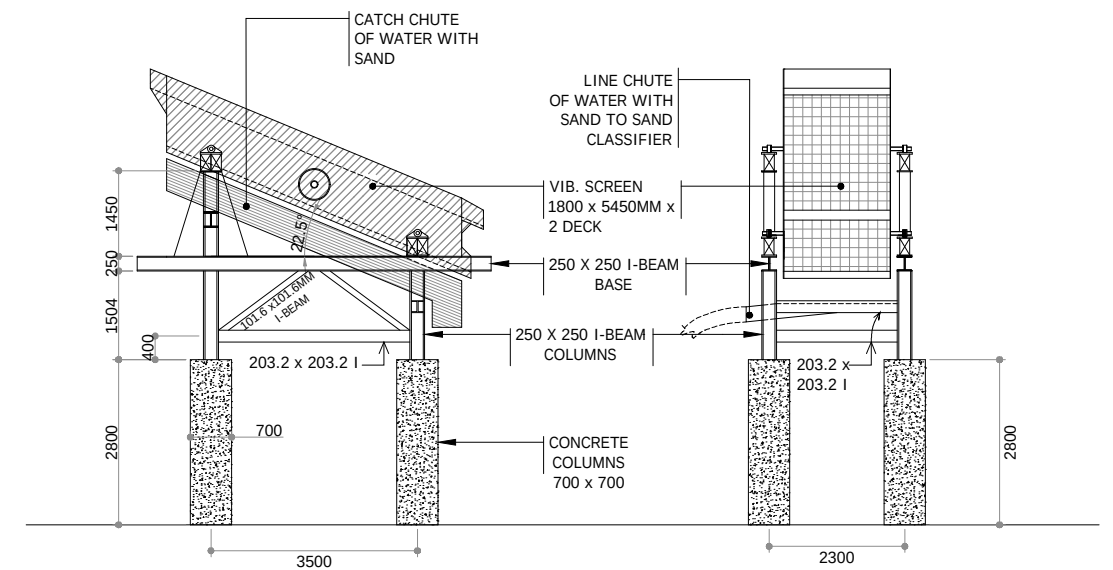
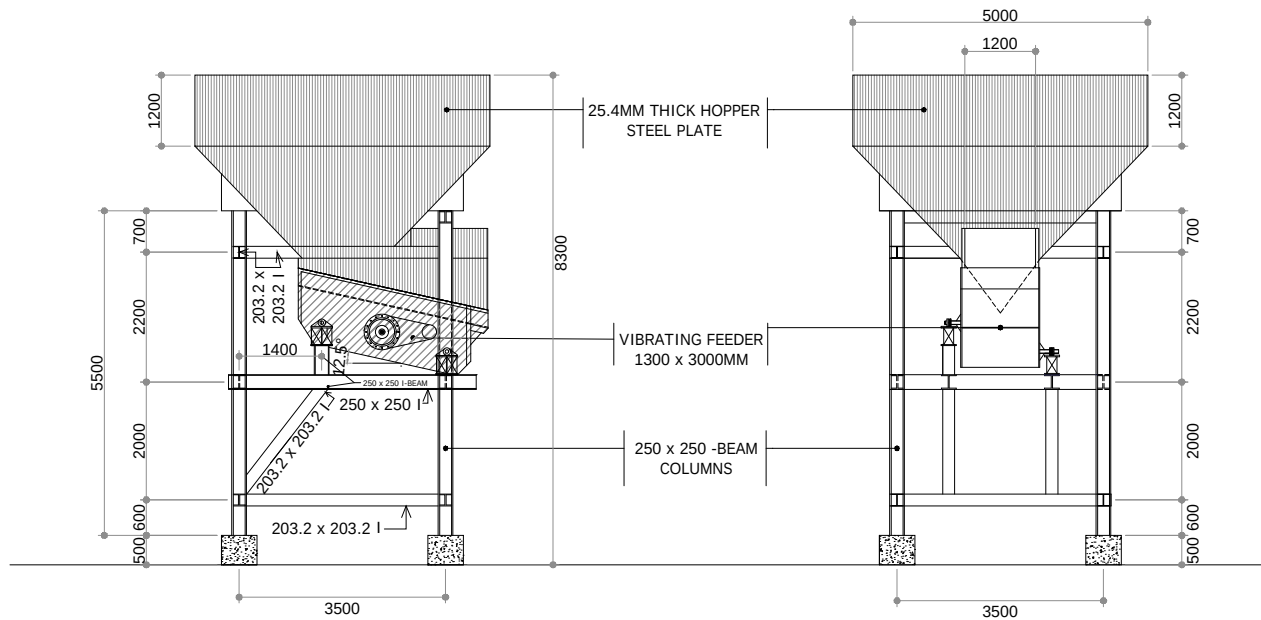
Project		FAB/ASSY.DRAWING		
- GRINDING CEMENT PLANT		Drawn By	D. SOMBILON	Checked By
Date		JANUARY - 2021	Scale	Paper Size A4
Drawing Title		CAD Reference		
- MILL 2 LINER 305 X 490 MM		Drawing #	Sheet 1/1	Revision



DETAILS OF VIBRATING FEEDER
1300MM X 3000MM LONG



DETAILS OF VIBRATING SCREEN
1.8 x 5.45 x 2DECK

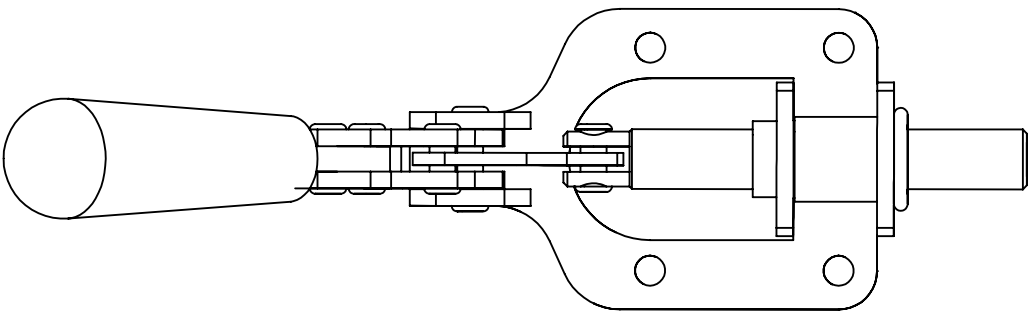


ENGINEER:	OWNER:	PROJECT TITLE:	SHEET CONTENTS:	SHEET NO.
	By: _____	ROCK CRUSHING PLANT	VIBRATING FEEDER AND VIBRATING SCREEN	A-1
		LOCATION	DETAILS AND SETUP	
			END DATE: _____ CHECKED BY: _____	

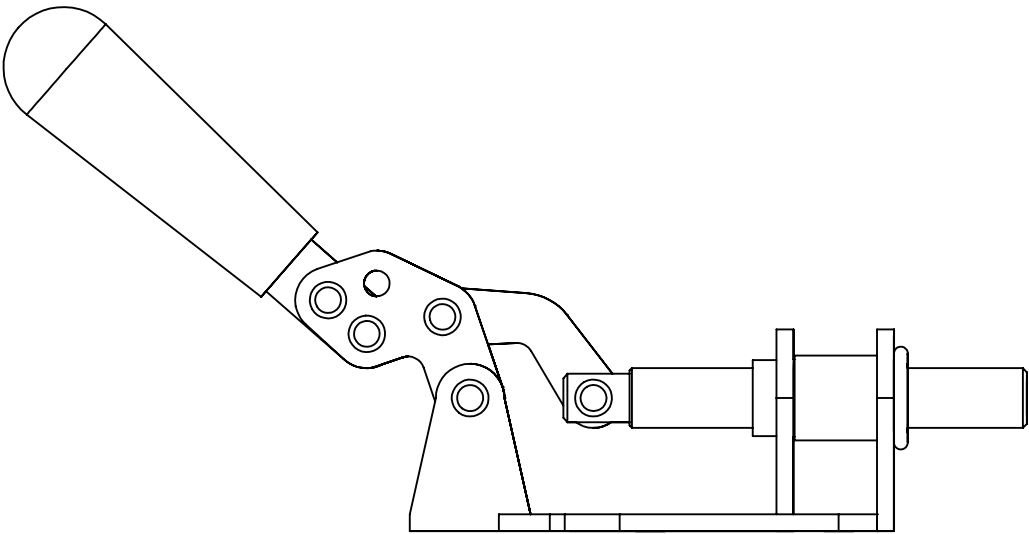
SOLIDWORKS

SOLID MODELLING

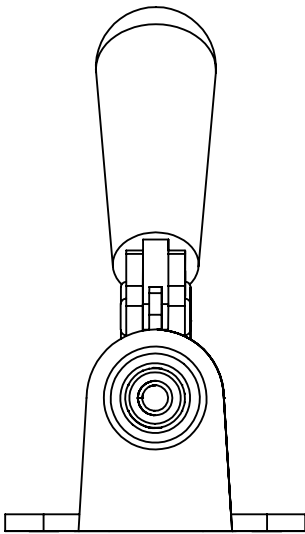
SAMPLE PORTFOLIO



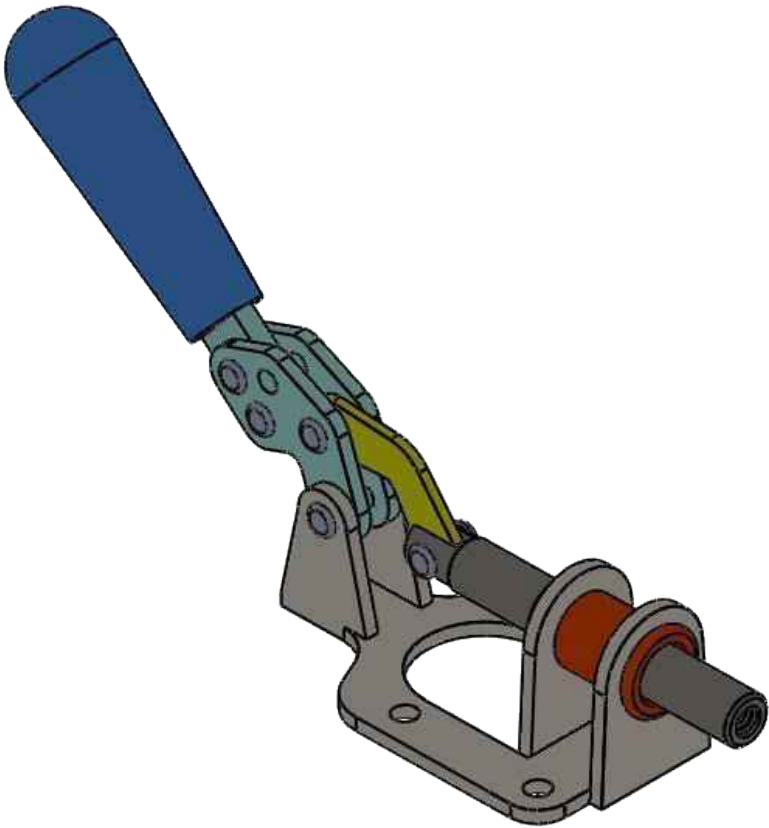
(1 : 1.4)
TOP VIEW



(1 : 1.4)
LEFT VIEW



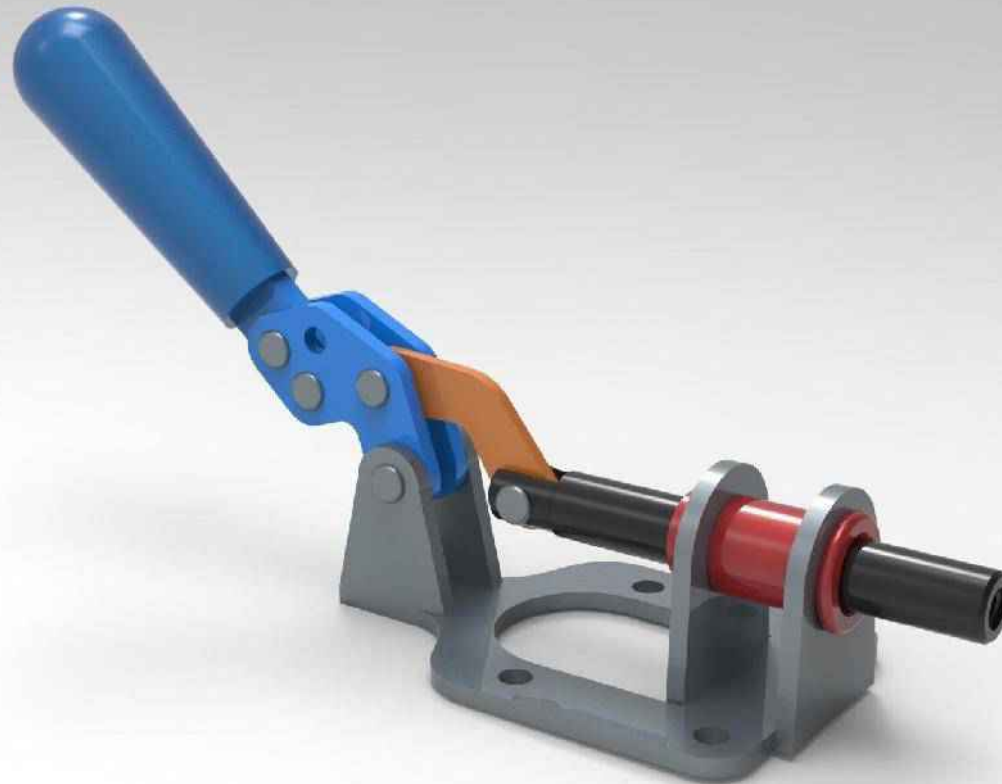
(1 : 1.4)
FRONT VIEW



(1 : 1.4)
ISOMETRIC

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:				FINISH:		DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING		REVISION
	NAME	SIGNATURE	DATE				TITLE: TOGGLE CLAMP ASSEMBLY		
DRAWN	DAREN								
CHK'D									
APPV'D									
MFG									
Q.A							DWG NO.		A3
							WEIGHT:		SHEET 1 OF 2
							SCALE:1:2		

TOGGLE CLAMP ASSEMBLY RENDERING

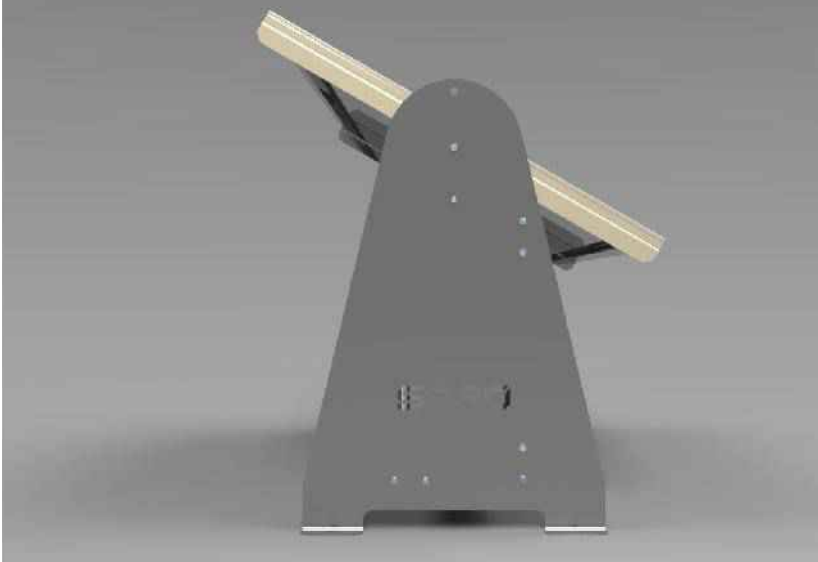
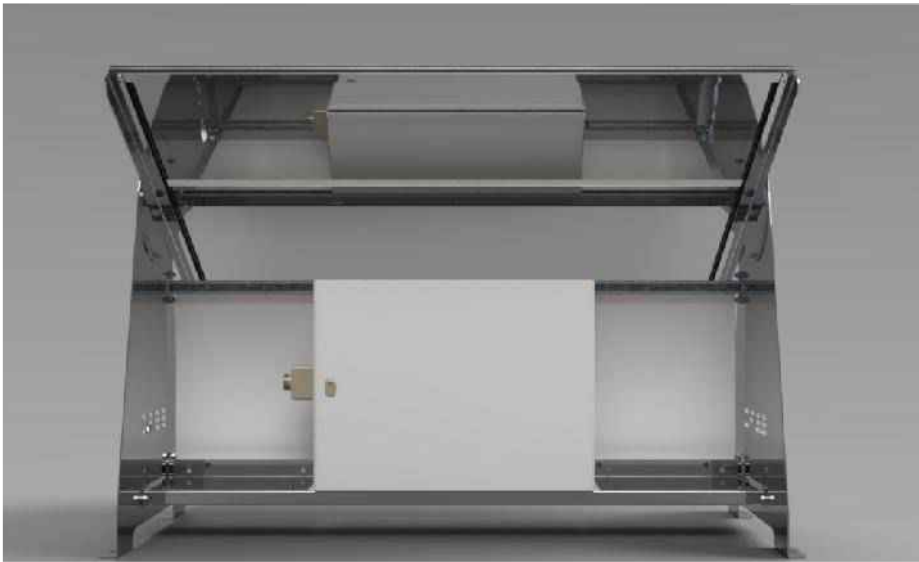


3D Rendering Samples from AutoCAD



Onshape 3D Model Rendering Sample





FABRICATED BY	DATE
FABRICATION CHECK BY	DATE
WELD CHECK BY	DATE

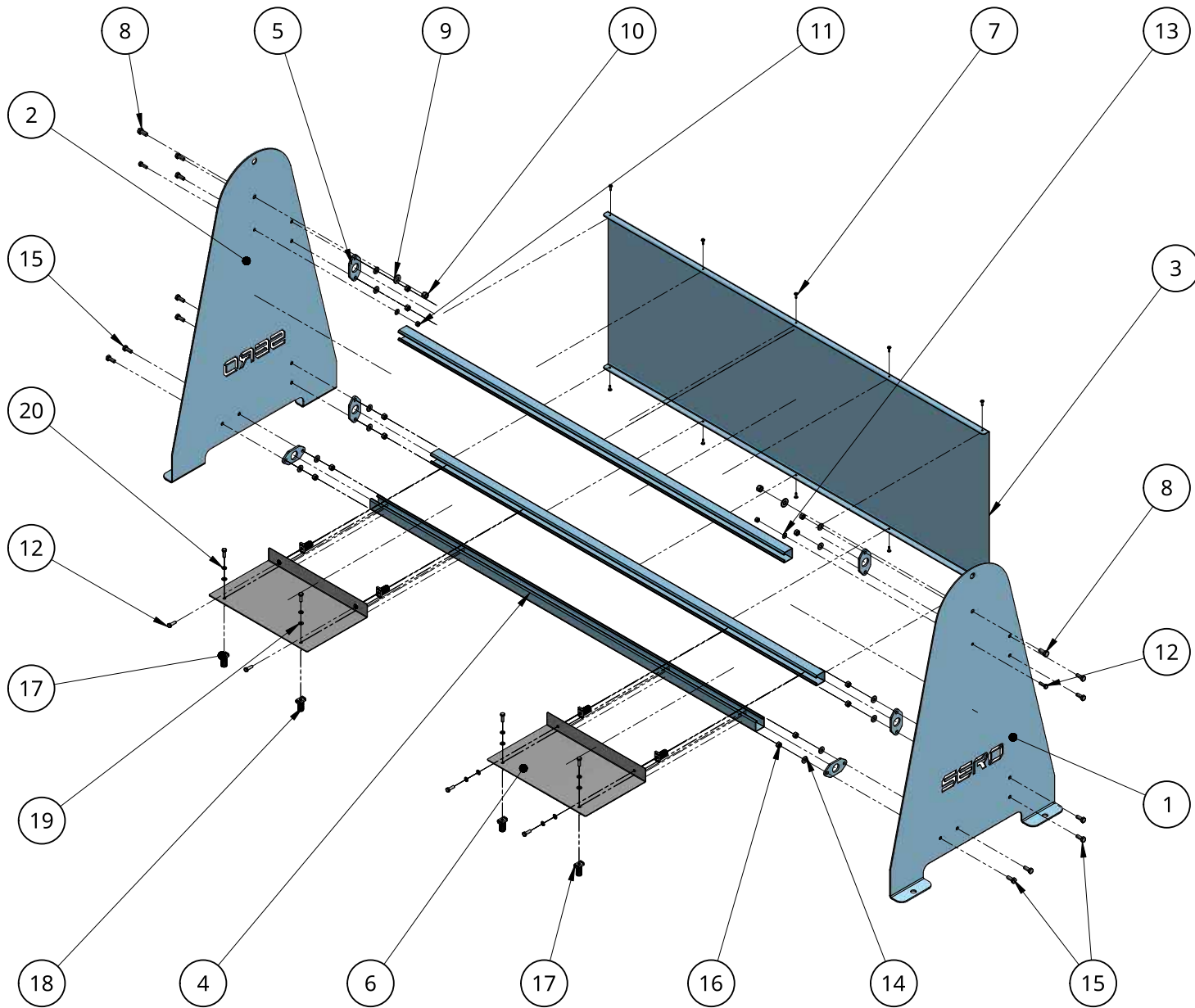
A	07/10/2024	ISSUE FOR APPROVAL	DS	
REV	DATE	DESCRIPTION	DRN	CHK

3D RENDERING

GENERAL NOTE

ALL STEELWORK TO BE FABRICATED & ERECTED TO AS4100
ALL WELDS TO BE 6MM C.F.W- U.N.O.
ALL WELDING TO AS1554
ALL CLEATS TO BE ON CENTERLINE U.N.O
HARDSTAMP ASSEMBLY MARK WHERE INDICATED THUS
ALL SNIPES ARE 15x45°

DRAWING TITLE				
CONTRACT				
MODELLED BY		CHECKED BY		
SHEET NO.	1 OF 1	DRAWING NO.		REVISION
SCALE.	1:10	-		A



ITEM	QTY	PART NUMBER	DESCRIPTION	DIMENSION
1	1	Z-PRT-RSC1A-00	Right Side Sero Cover	1165 x 750 x 4mmx GR250
2	1	Z-PRT-LSC1A-00	Left Side Sero Cover	1165 x 750 x 4mmx GR250
3	1	Z-PRT-BCP1A-00	Back Cover Plate	1740 x 655 x 2mm PL
4	3	Z-PRT-UNS1A-00	P100T Unistrut	1758 x 41.28
5	6	Z-PRT-CNP1A-00	Connecting Plate	100 x 49 x 6mm GR250
6	2	Z-PRT-MUT1A-00	Mounting Trays	324.5 x 450 x PGI
7	10	Z-ACC-SDS1A-00	Self Drilling Screw	ST4.8 x 16mm
8	2	Z-ACC-HEX12A-00	Hex Head Screw	M12 x 25 Galvanised Steel
9	3	Z-ACC-WSH12A-00	Washer	M12 x 25 Galvanised Steel
10	2	Z-PRT-NUT12A-00	Nylock Nut	M12 x 25 Galvanised Steel
11	2	Z-PRT-NUT8A-00	Nylock Nut	M8x 20 Galvanised Steel
12	10	Z-ACC-HEX8A-00	Hex Head Screw	M8x 20 Galvanised Steel
13	2	Z-ACC-WSH8A-00	Washer	M8x 20 Galvanised Steel
14	12	Z-ACC-WSH10A-00	Washer	M10x 25 Galvanised Steel
15	12	Z-ACC-HEX10A-00	Hex Head Screw	M10x 20 Galvanised Steel
16	12	Z-ACC-NUT10A-00	Nylock Nut	M10x 20 Galvanised Steel
17	8	Z-ACC-SPN8A-00	Spring Nut	M8x 25 Galvanised Steel
18	8	Z-ACC-SPR8A-00	Spring	M8x 25 Galvanised Steel
19	8	Z-ACC-FLW8A-00	Flat Washer	M8x 25 Galvanised Steel
20	8	Z-ACC-SPW8A-00	Spring Washer	M8x 25 Galvanised Steel

FABRICATED BY	DATE
FABRICATION CHECK BY	DATE
WELD CHECK BY	DATE

A	07/08/2024	ISSUE FOR APPROVAL	DS	
REV	DATE	DESCRIPTION	DRN	CHK

DRAWING : ASSEMBLY

GENERAL NOTE
ALL STEELWORK TO BE FABRICATED & ERECTED TO AS4100
ALL WELDS TO BE 6MM C.F.W- U.N.O.
ALL WELDING TO AS1554
ALL CLEATS TO BE ON CENTERLINE U.N.O
HARDSTAMP ASSEMBLY MARK WHERE INDICATED THUS
ALL SNIPES ARE 15x45°

DRAWING TITLE	DRAWING ASSEMBLY			
CONTRACT				
MODELLED BY		CHECKED BY		
SHEET NO.	1 OF 1	DRAWING NO.		REVISION
SCALE.	1:10	-		A